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(54) ROLLING COLLAPSIBLE TRAVEL LUGGAGE

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(21) Appl. No.: **19/213,665**

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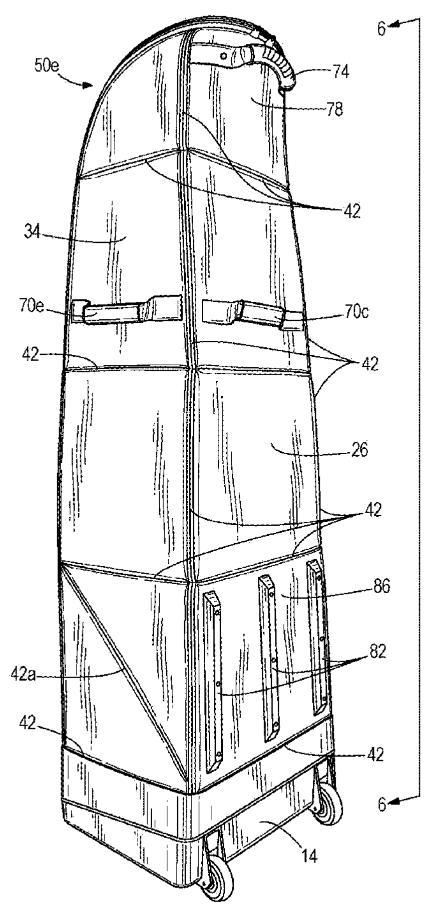
CPC ***A45C 7/0036*** (2013.01); ***A45C 5/14*** (2013.01); *A45C 13/002* (2013.01); *A45C 13/103* (2013.01)

(57) **ABSTRACT**

A rolling luggage bag includes a cover fixedly coupled to a base having a first side opposite a second side and a bottom face extending therebetween. The cover comprises a plurality of panels defining an interior chamber, a first and second wheel coupled to the base and configured to rotate about an axis of rotation, at least a portion of each of the first and second wheels projecting from the first side and from the bottom face, and a third wheel and a fourth wheel coupled to the bottom face. The third and fourth wheels independently swivel about a respective swivel axis. When the rolling luggage is in an upright position, the first, second, third, and fourth wheels all contact a surface on which the luggage bag stands.



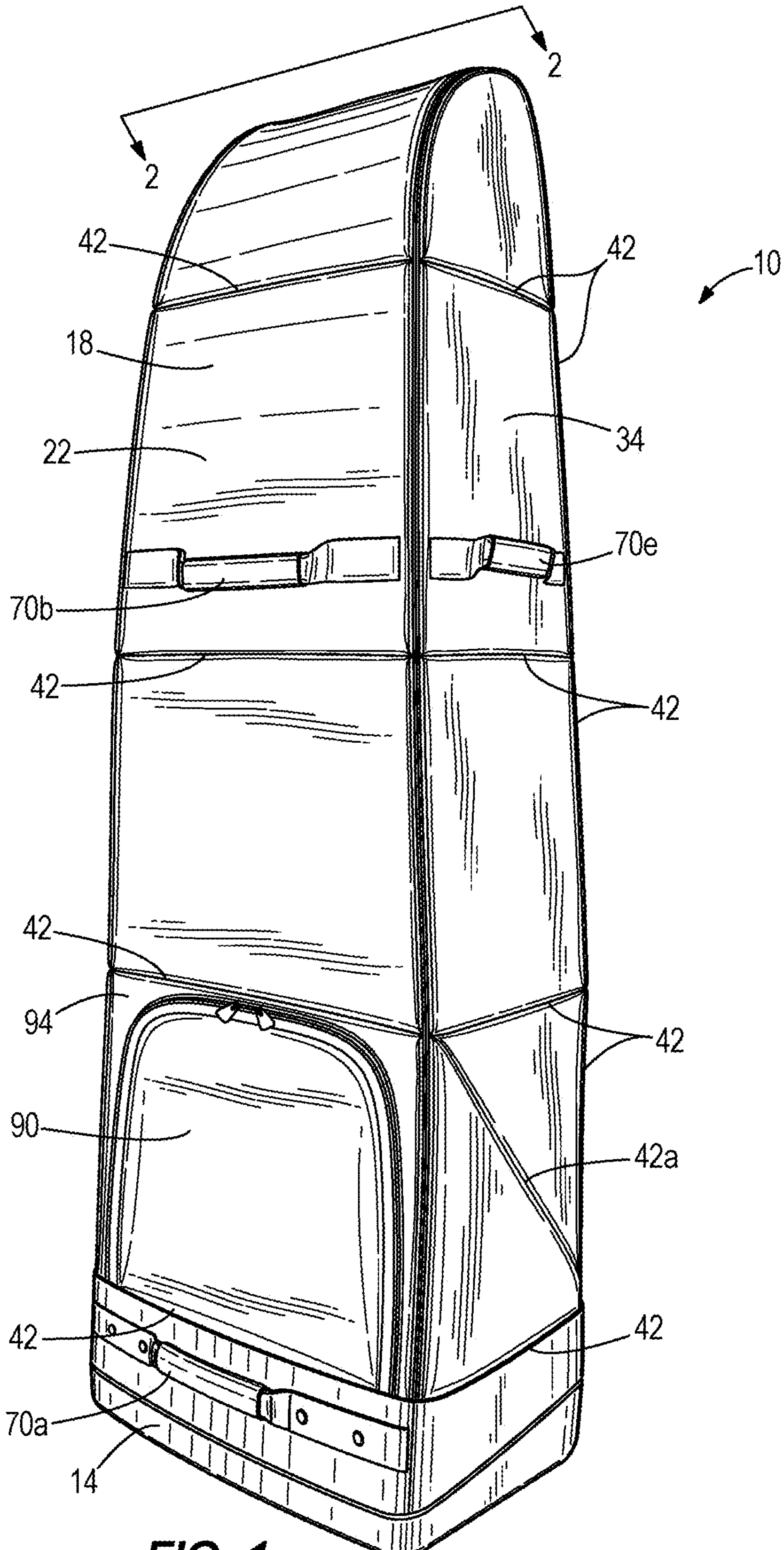


FIG. 1

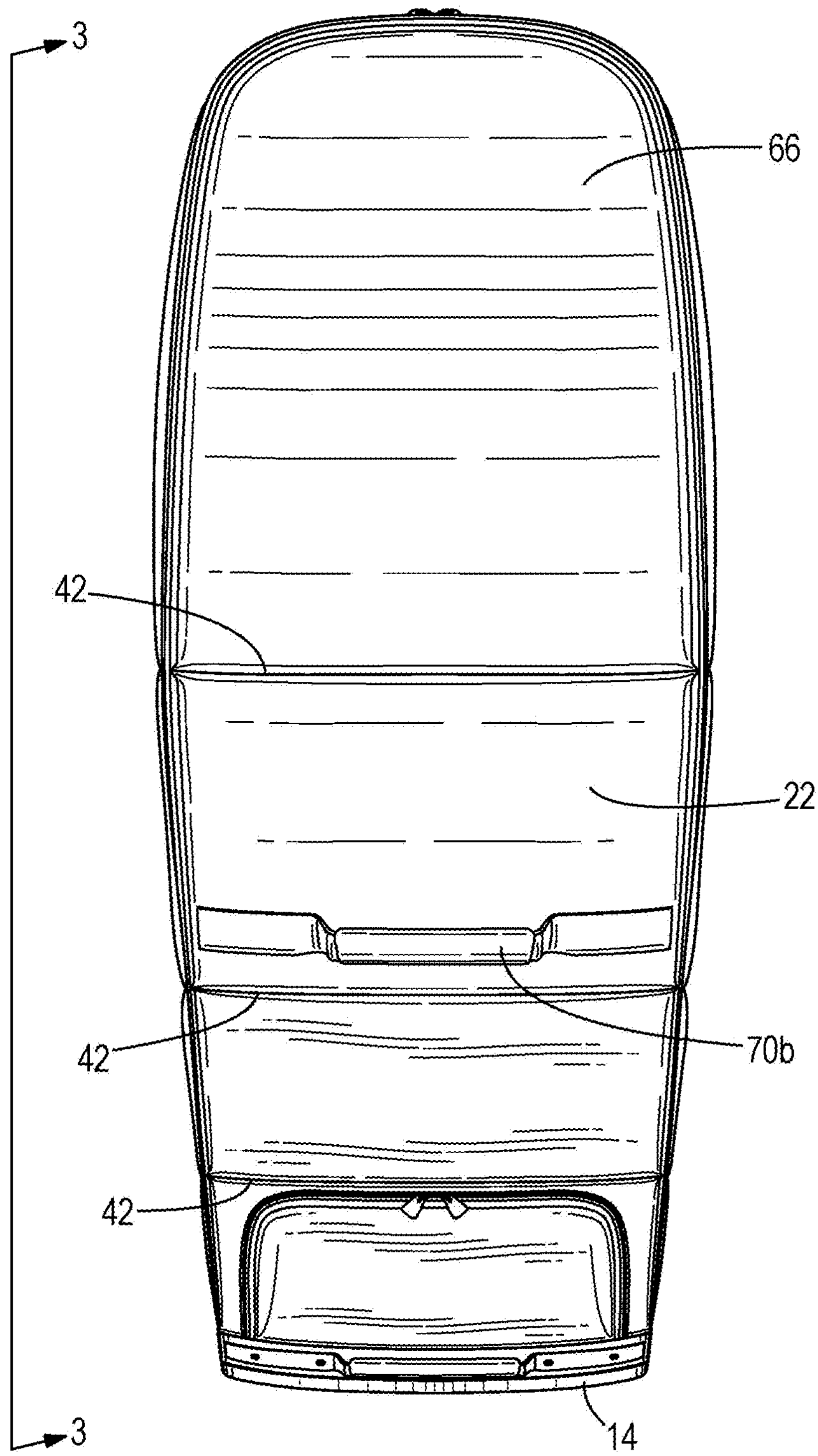
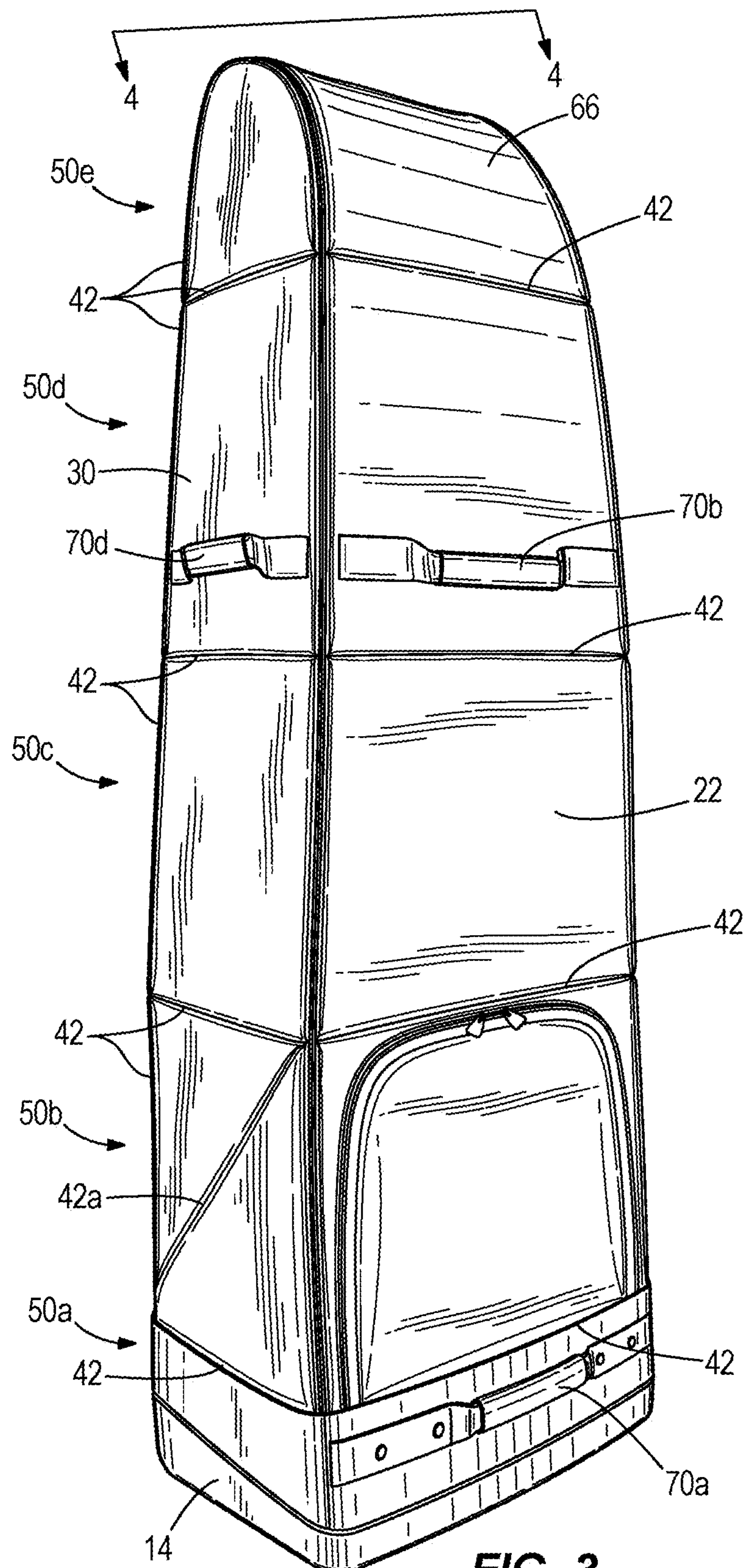


FIG. 2


4
4
66
50e
42
42
50d
30
70b
70d
42
42
50c
22
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42
70a
14

FIG. 3

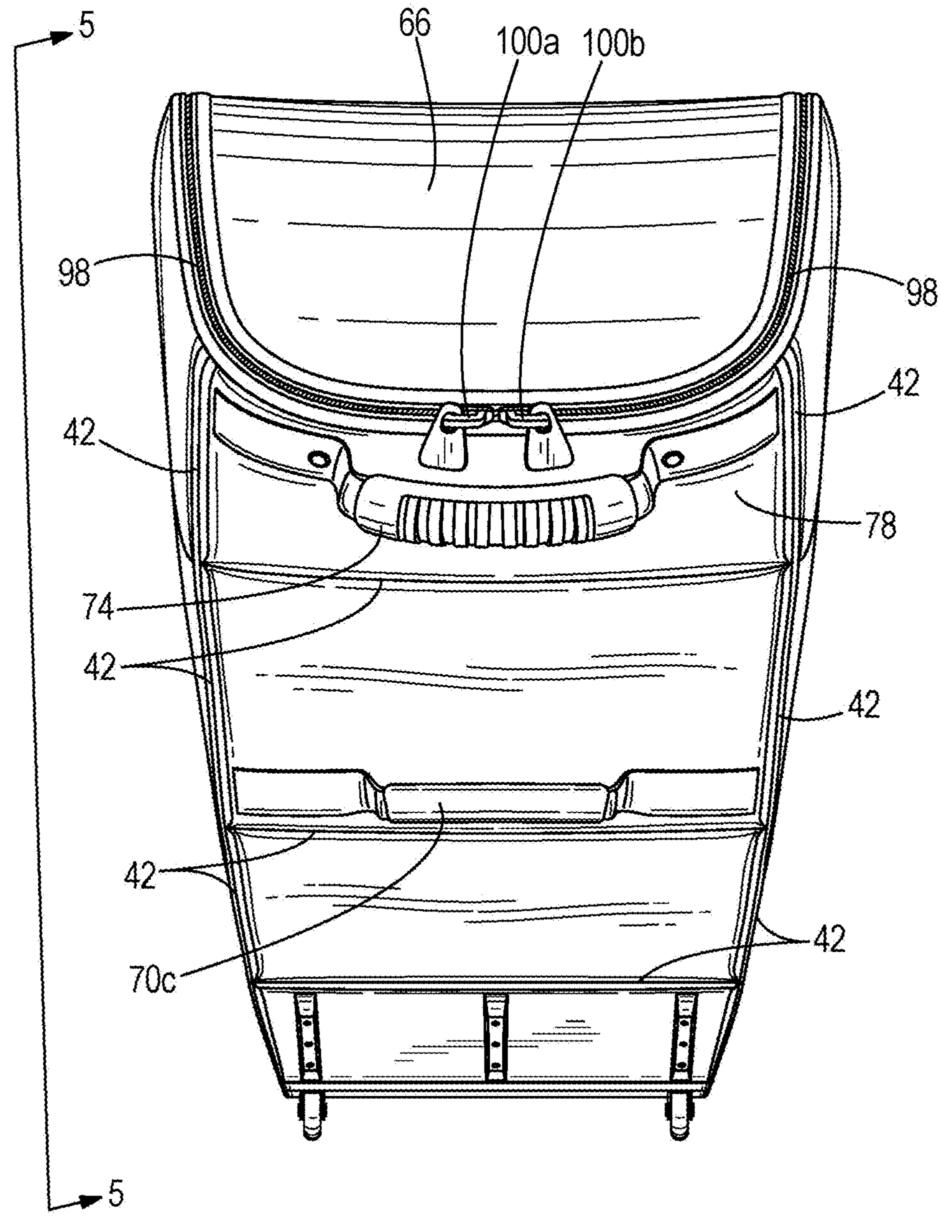


FIG. 4A

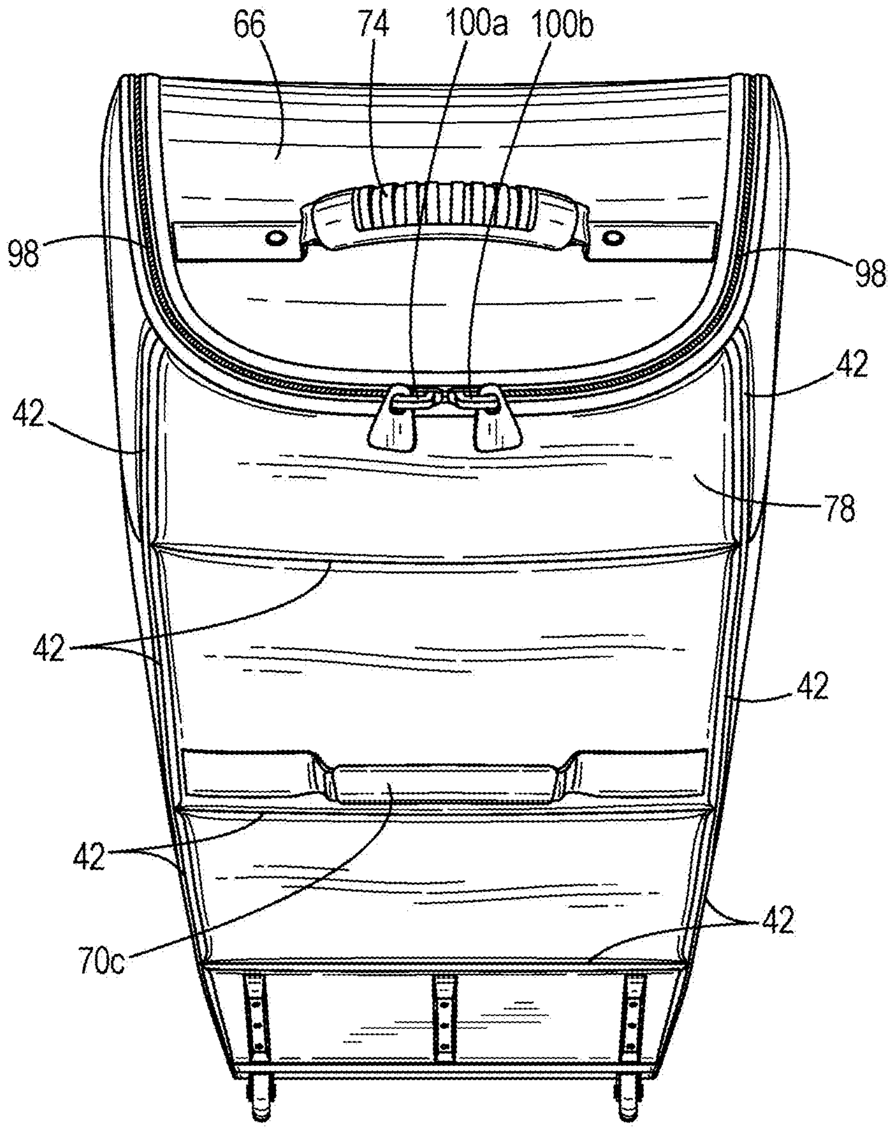


FIG. 4B

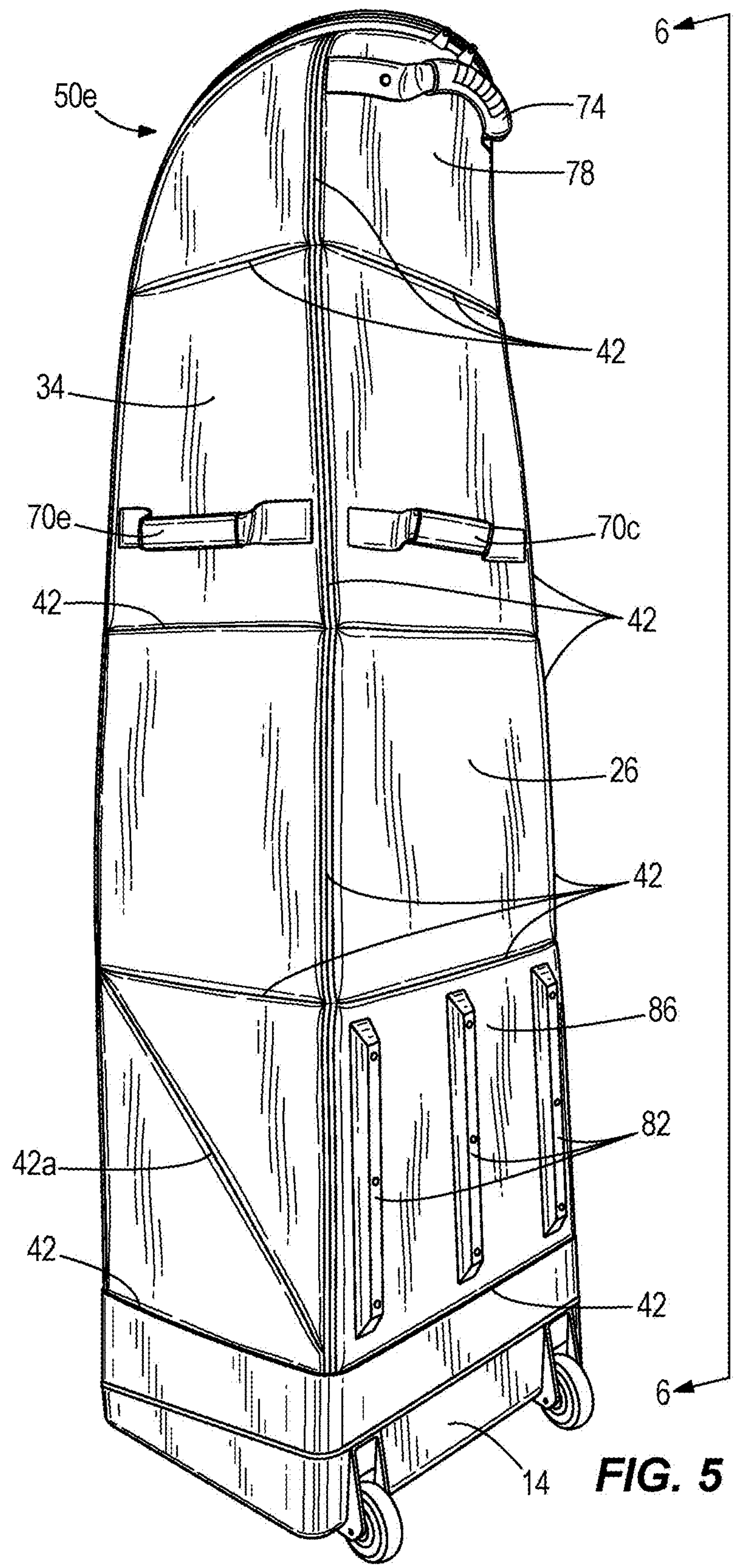


FIG. 5

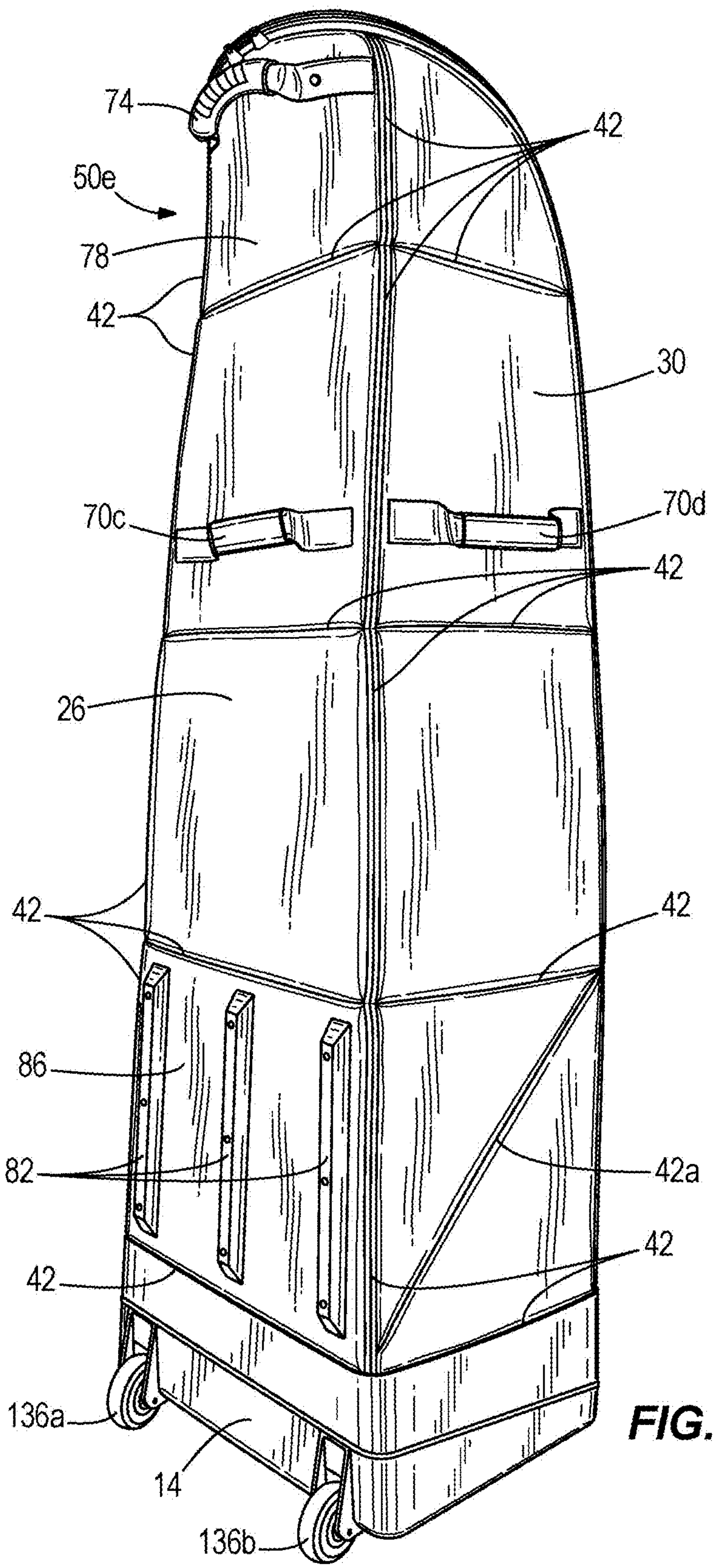
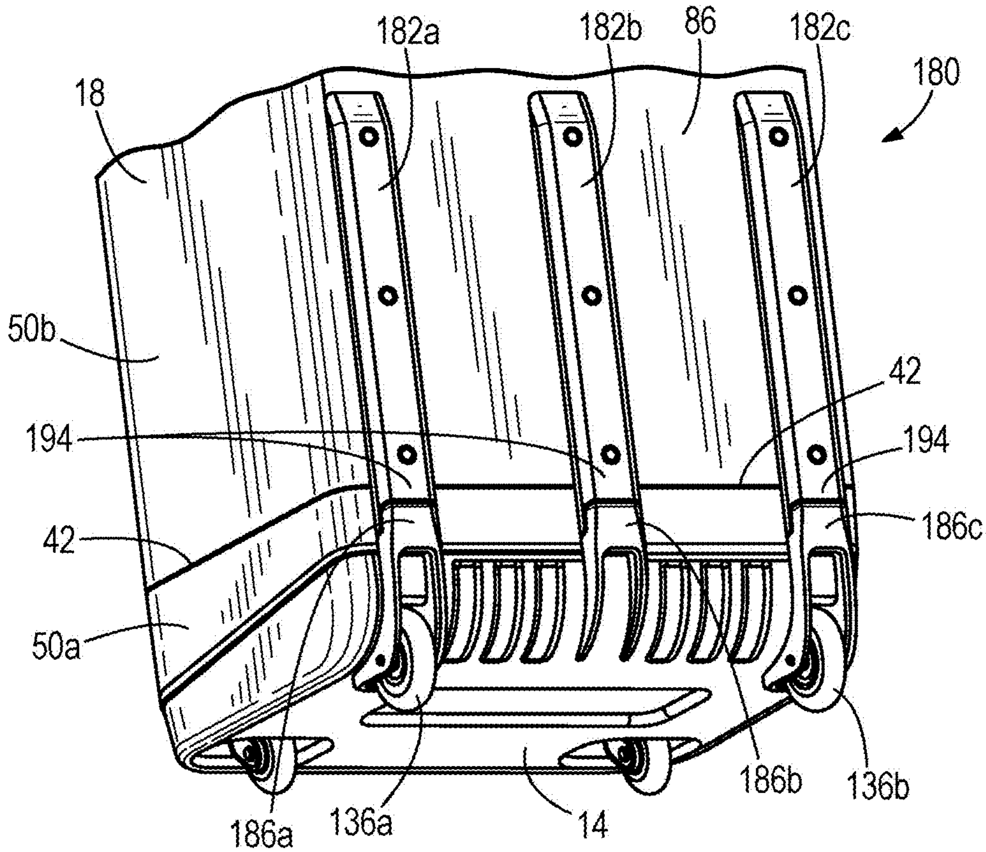


FIG. 6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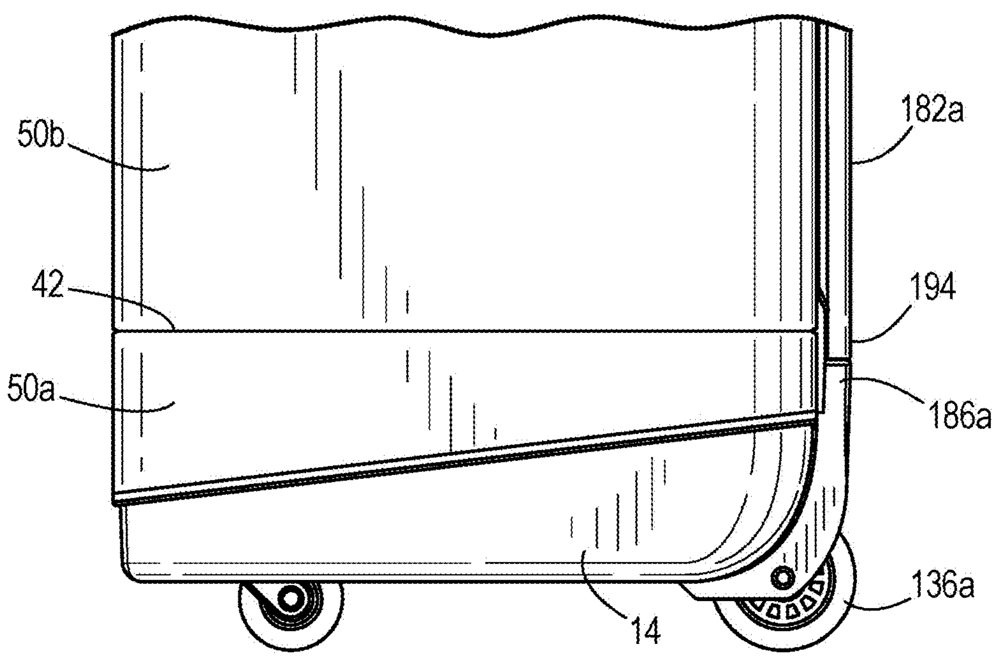
182b
86
182c
180
18
50b
42
194
194
186c
42
50a
186b
136b
186a
136a
14

FIG. 6A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182a
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194
50a
186a
136a
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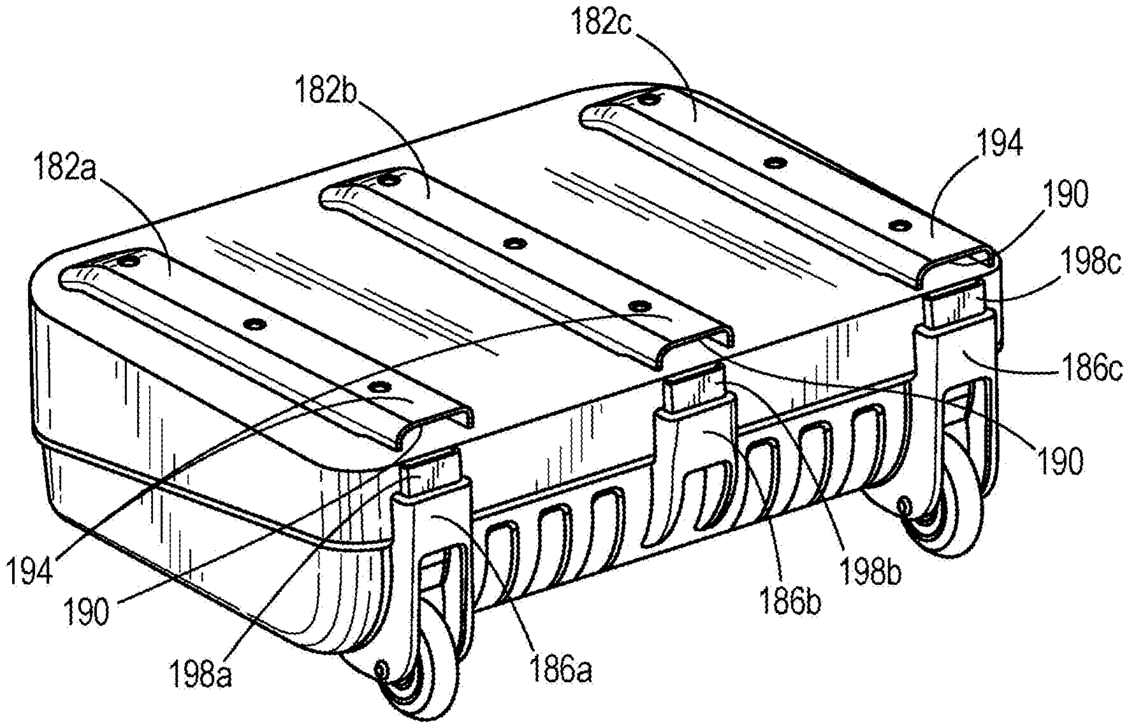
FIG. 6B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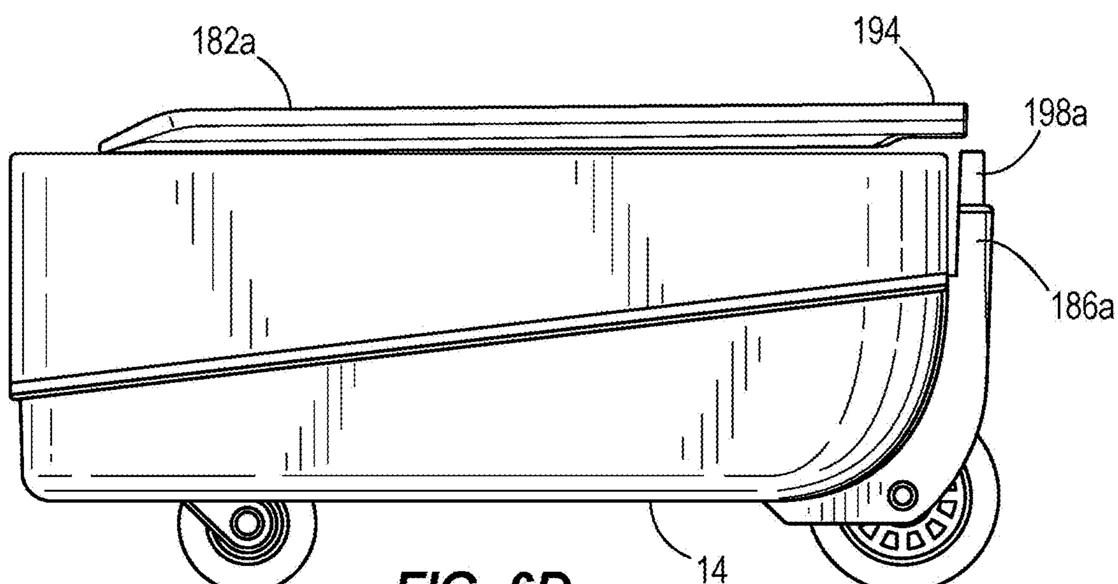
194
190
198a
186a
186b
198b

FIG. 6C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198a
186a
14

FIG. 6D

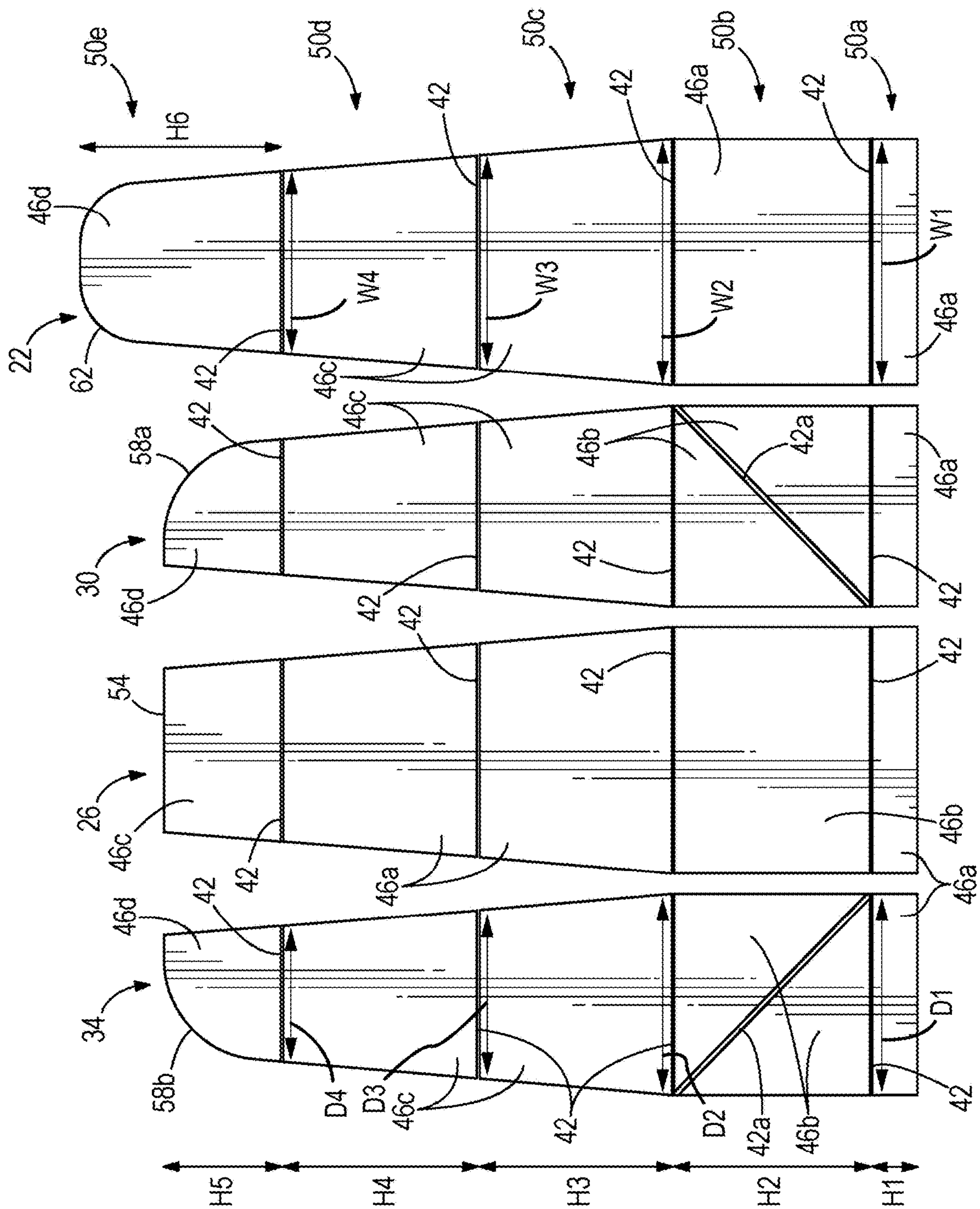
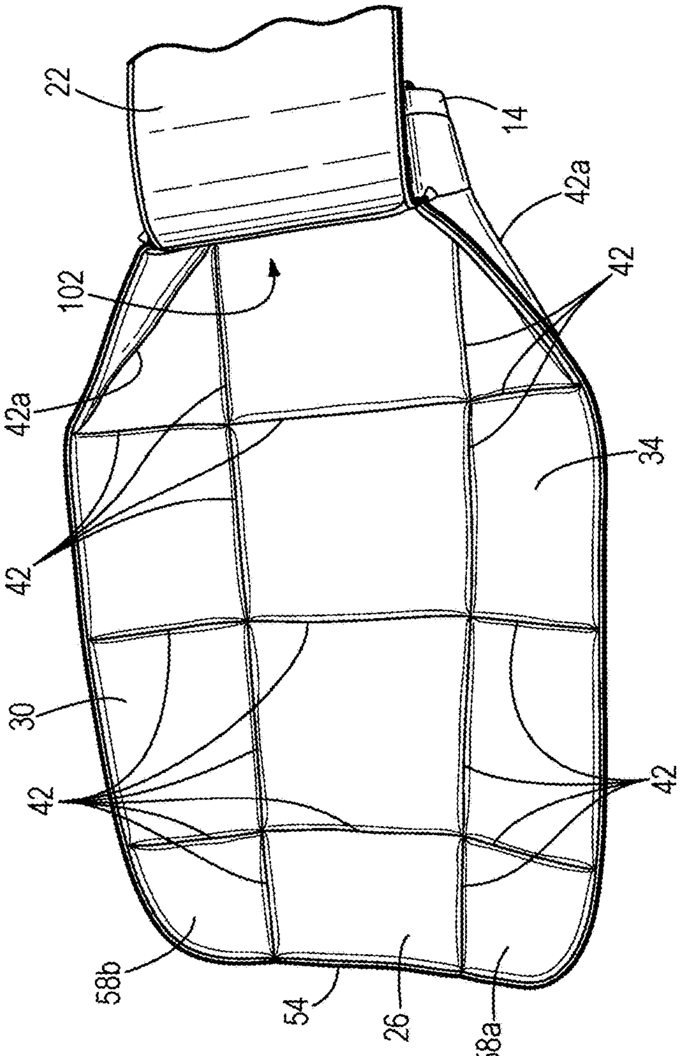


FIG. 7A FIG. 7B FIG. 7C FIG. 7D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42a
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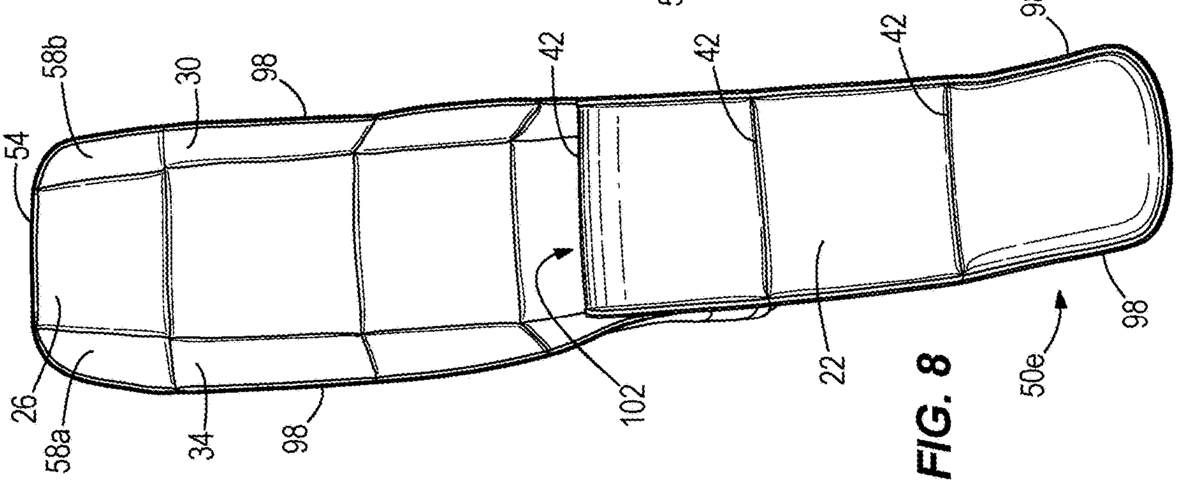
42
30
42
42
58b
54
26
58a

FIG. 9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42
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26
58a
34
98
102
22
98
50e

FIG. 8

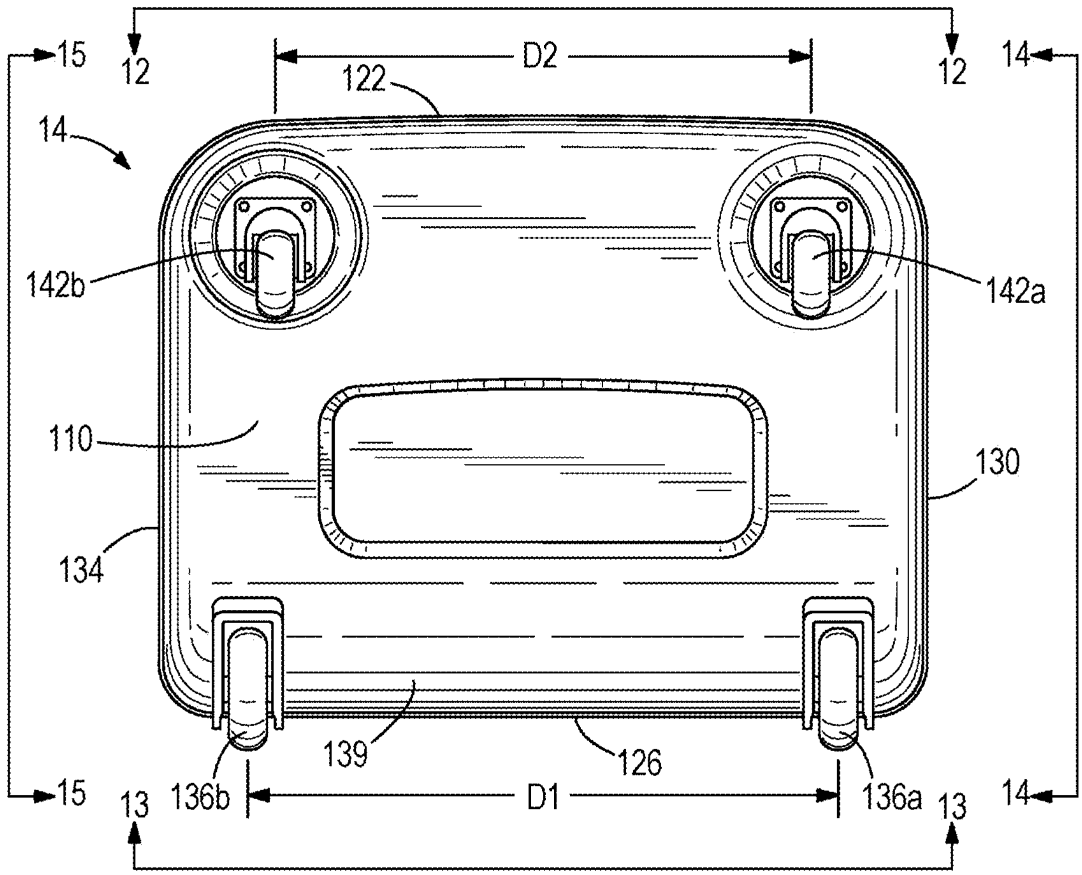
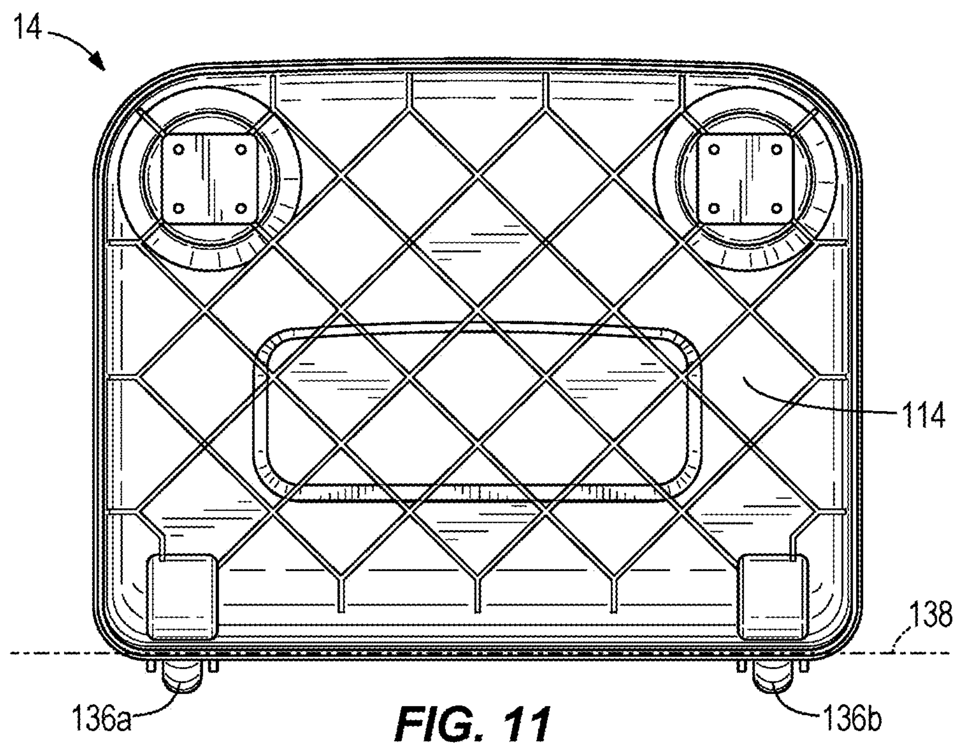

15
12
D2
12
14
122
14
142b
142a
110
130
134
139
126
15
13
136b
D1
136a
13
14

FIG. 10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136a
136b

FIG. 11

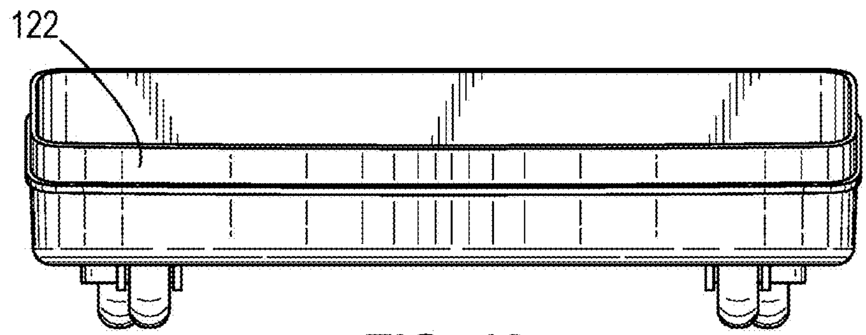


FIG. 12

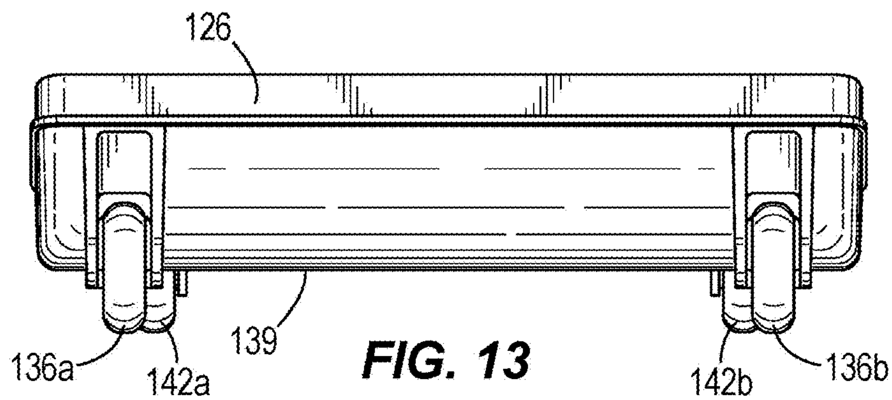


FIG. 13

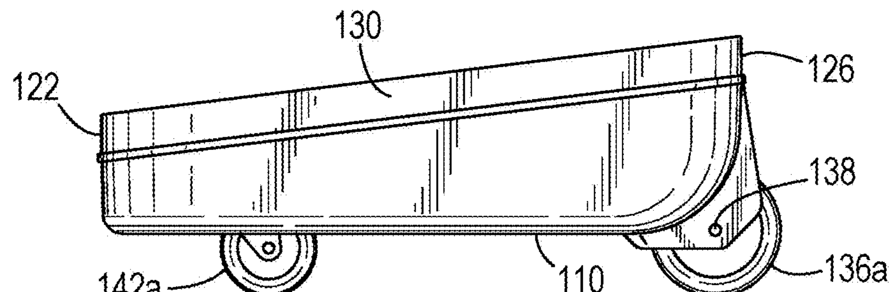


FIG. 14

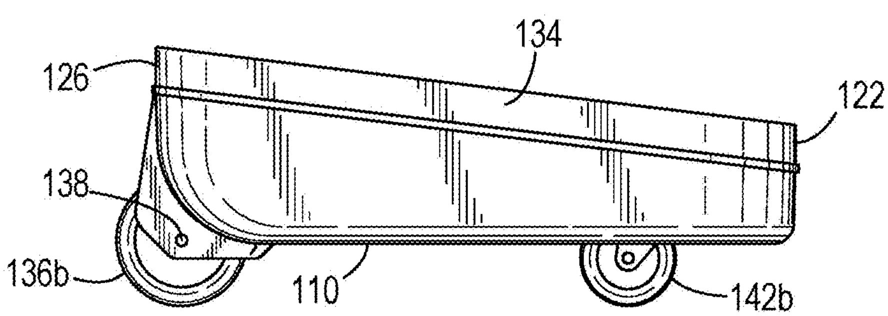


FIG. 15

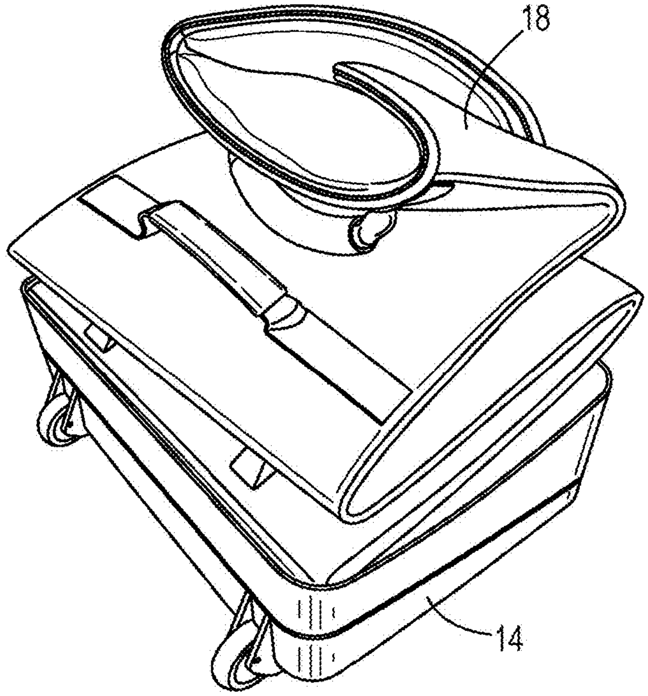


FIG. 17

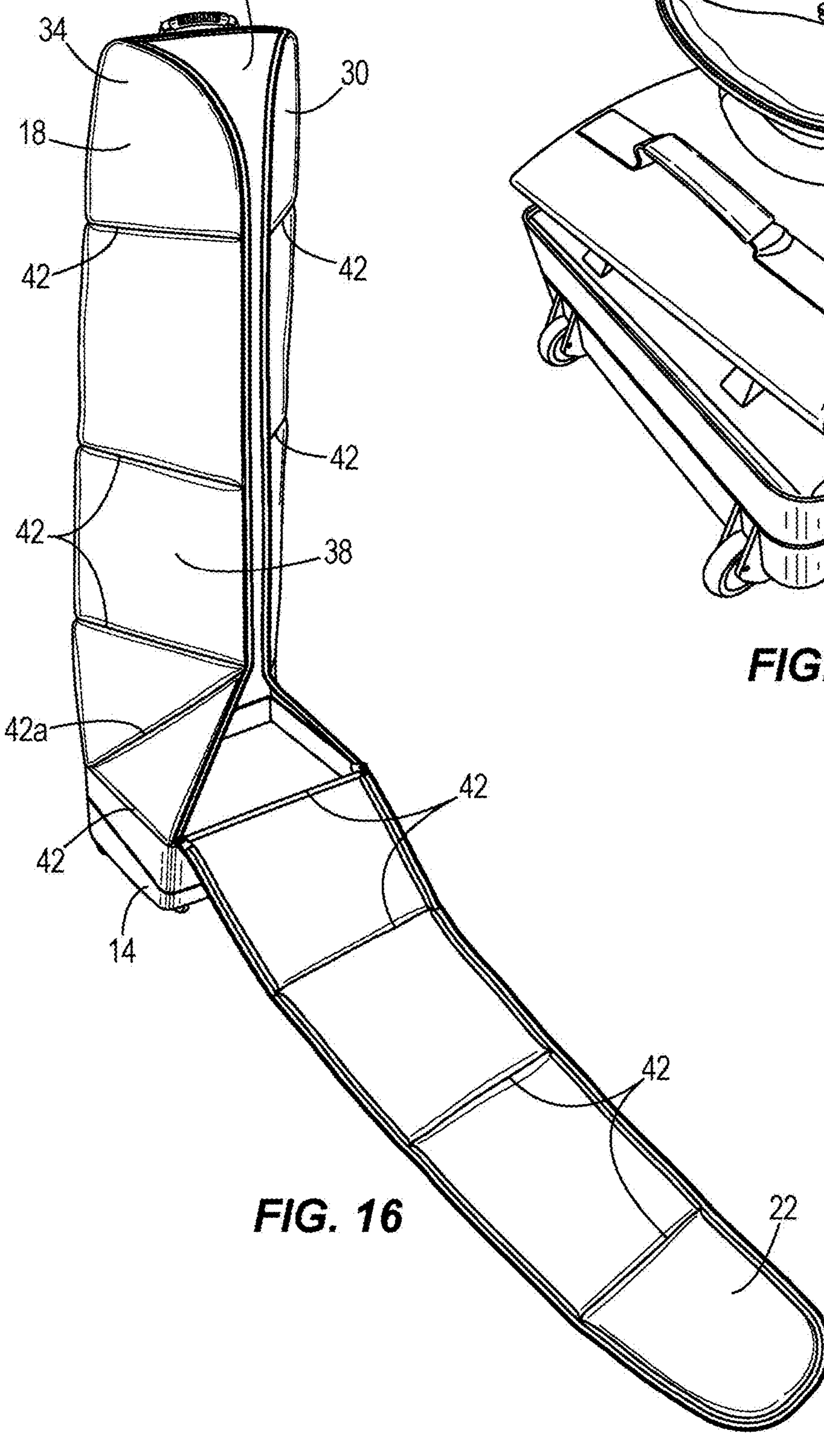


FIG. 16

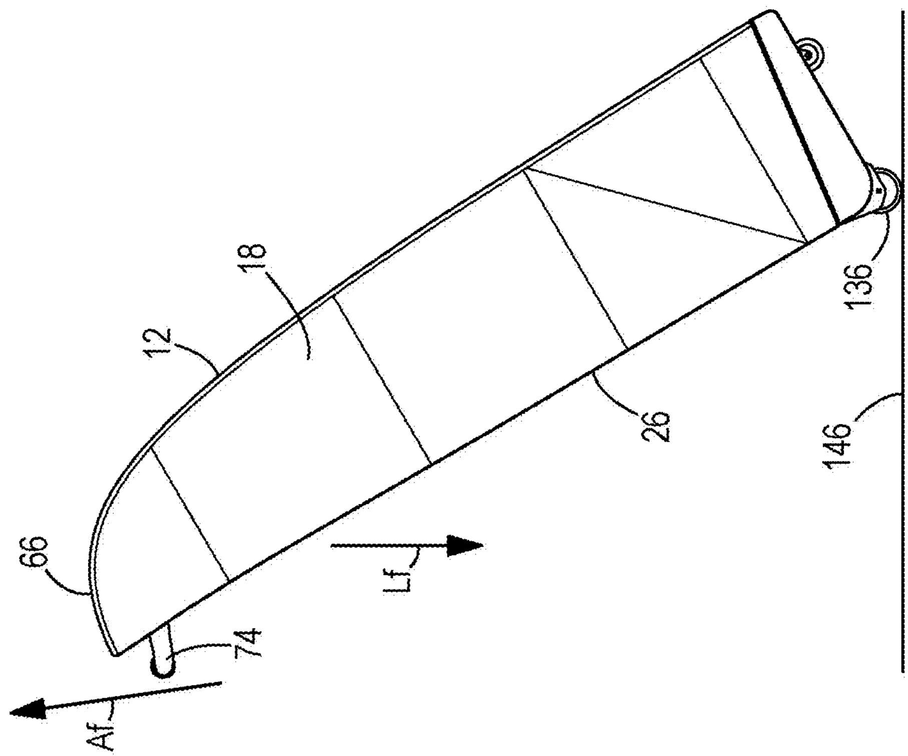


FIG. 19

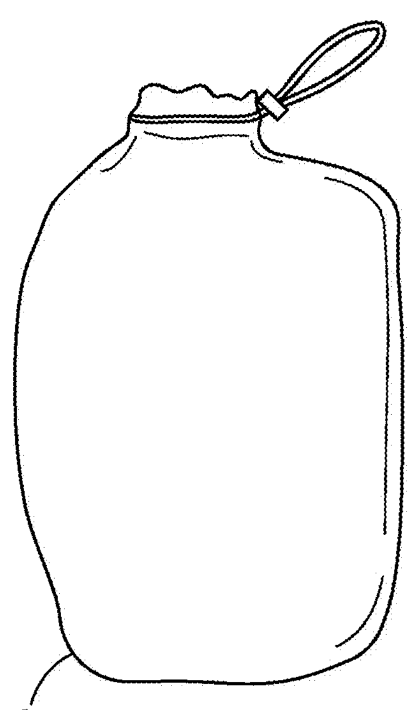


FIG. 18A

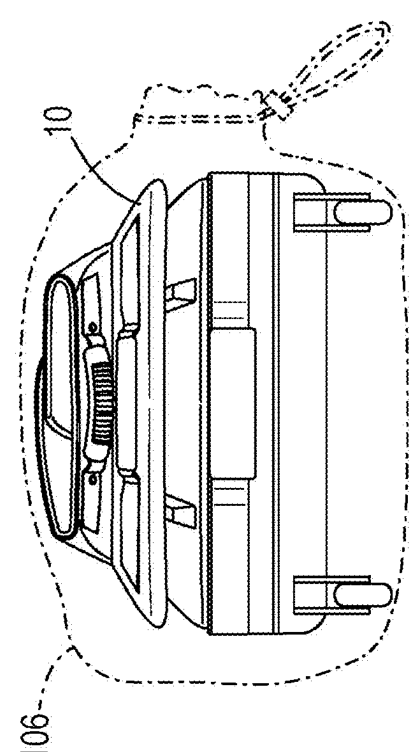
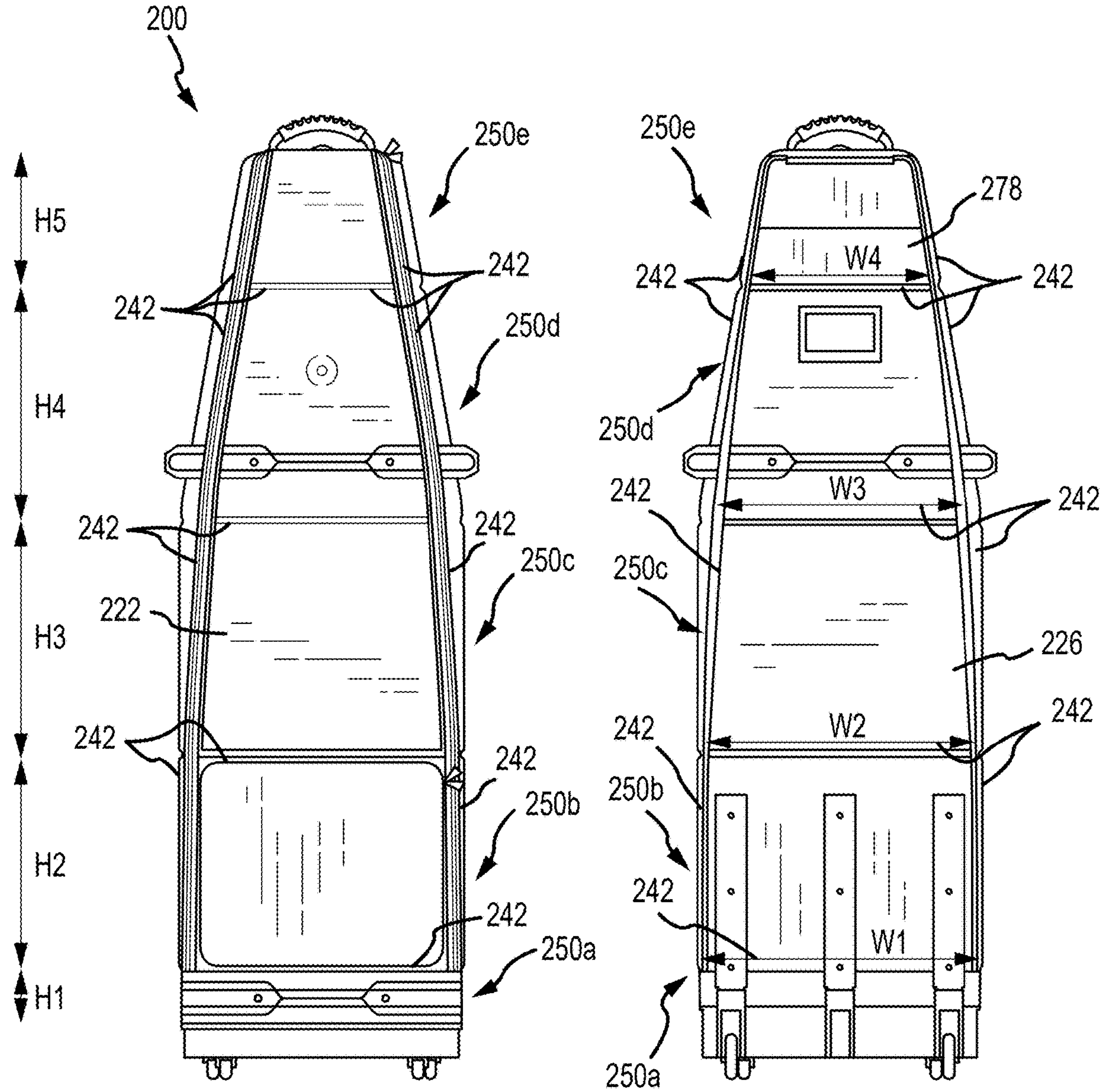


FIG. 18B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242
242
250d
H5
H4
H3
H2
H1
242
242
250c
222
242
242
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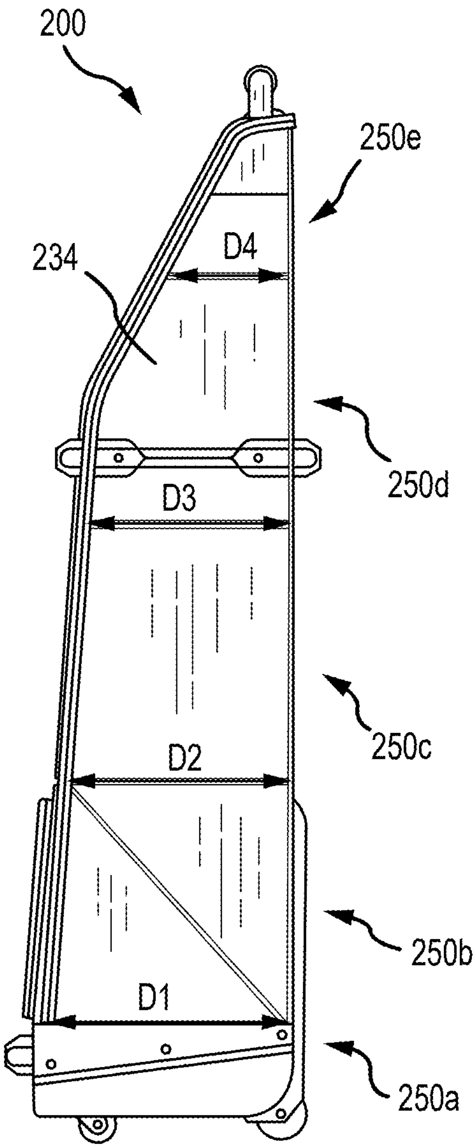
242
250a
250e
278
W4
242
242
250d
W3
242
242
250c
226
W2
242
242
250b
242
W1
250a

FIG.20

FIG.21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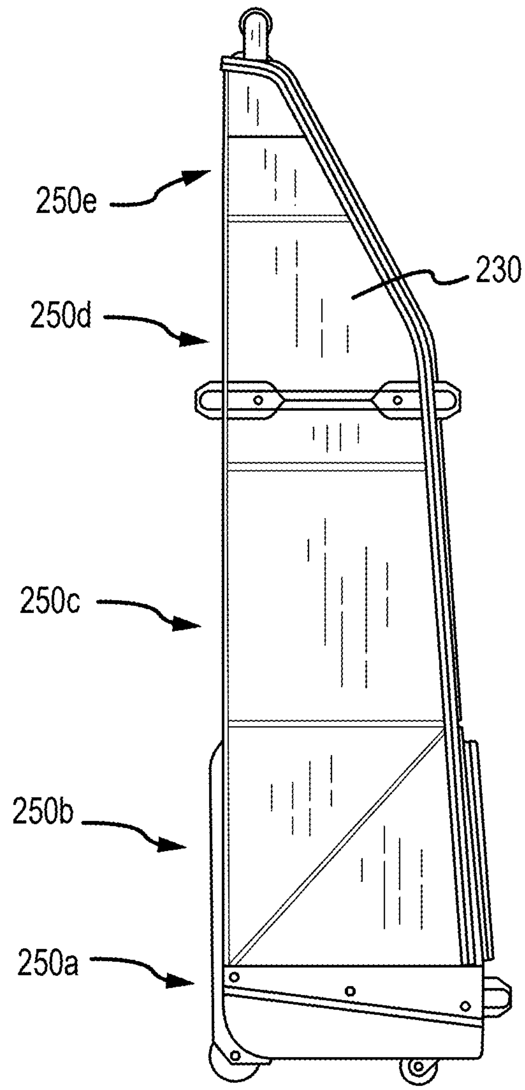
D3
250c
D2
250b
D1
250a

FIG.22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250b
250a

FIG.23

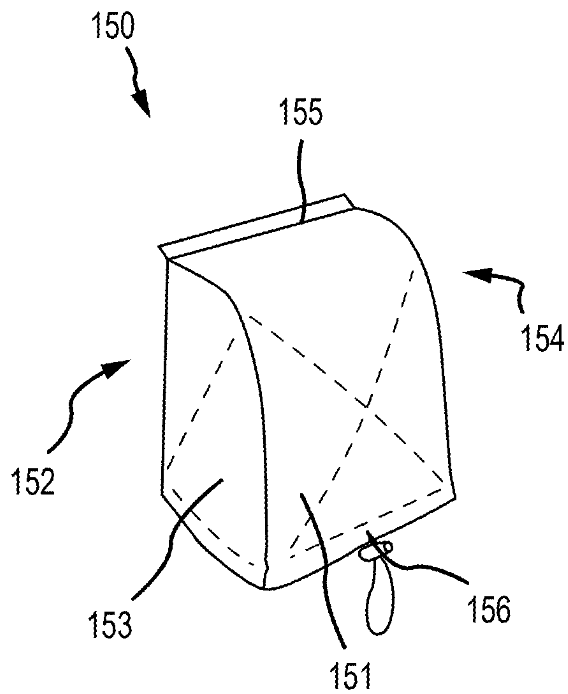


FIG.24

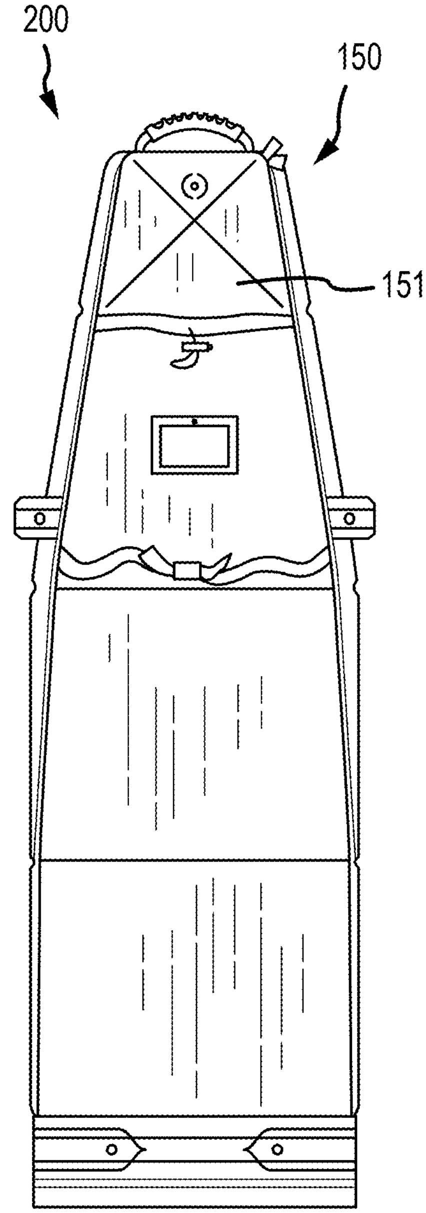


FIG.25

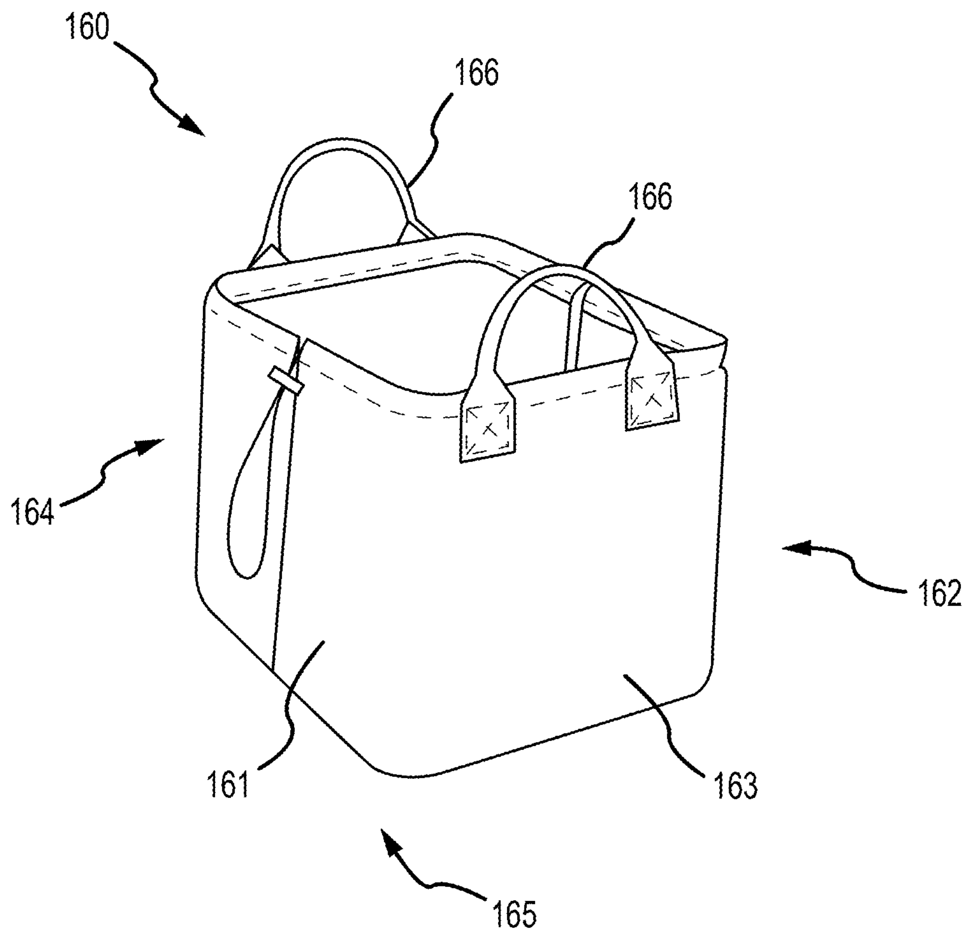
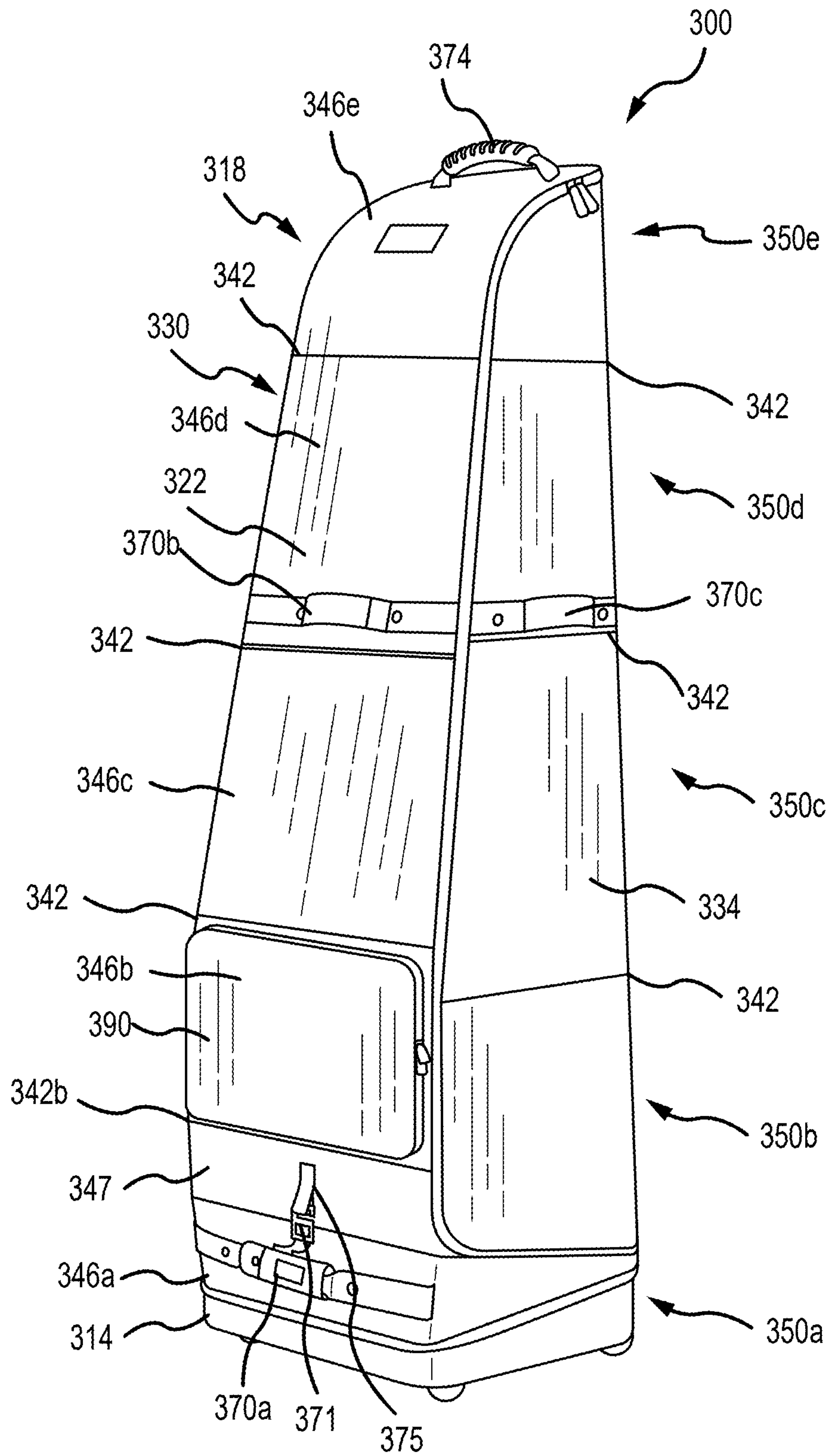


FIG.26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330
342
346d
350d
322
370b
370c
342
342
346c
350c
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346b
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390
342b
350b
347
346a
350a
314
370a
371
375

FIG.27A

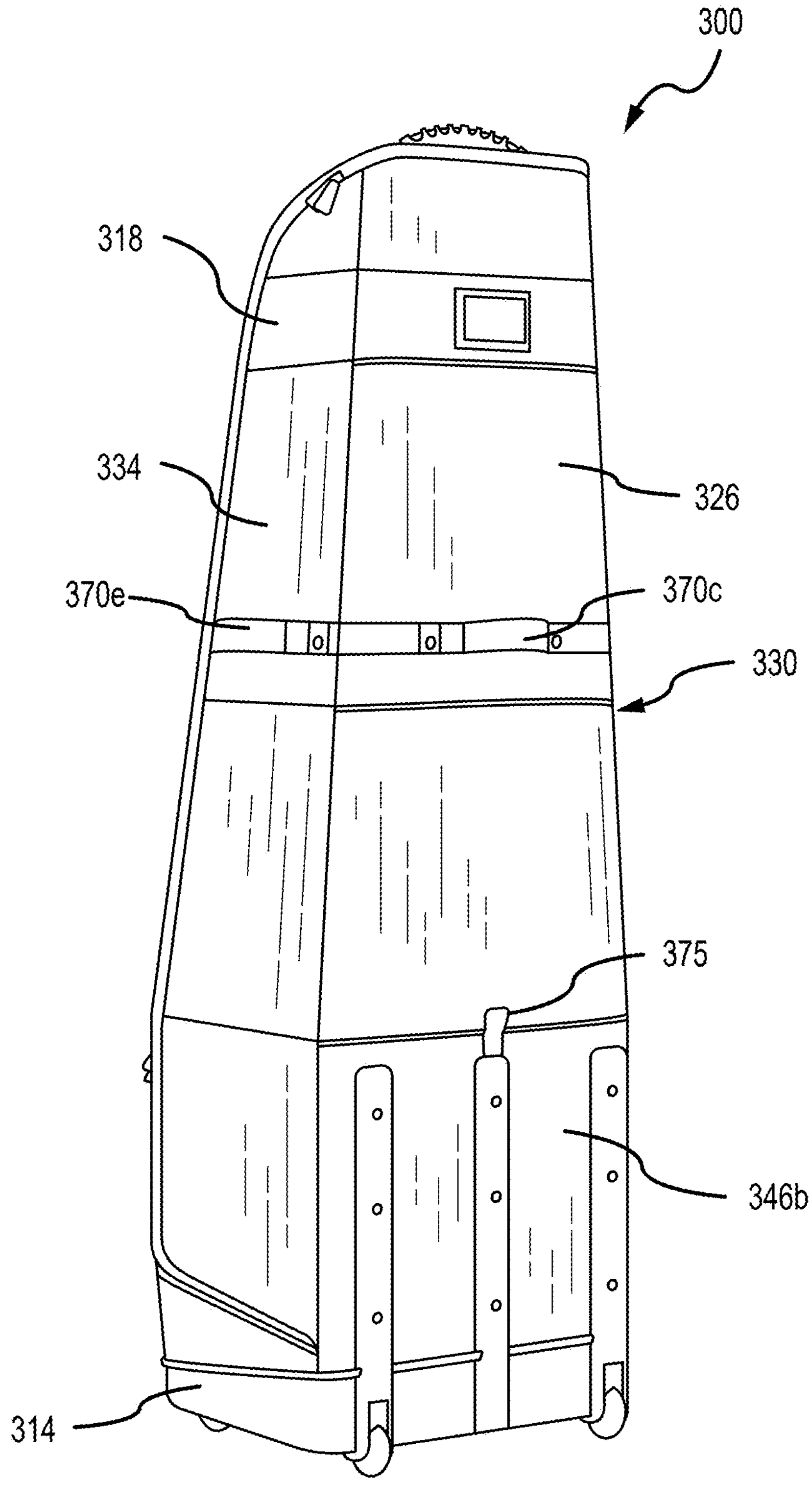

FIG.27B

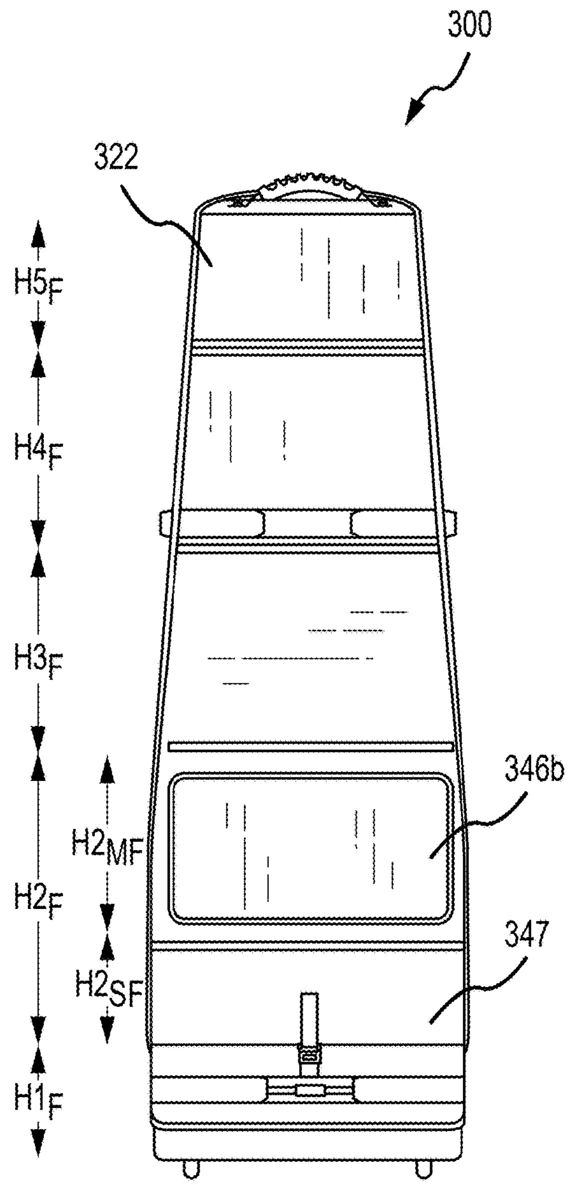

FIG.28A

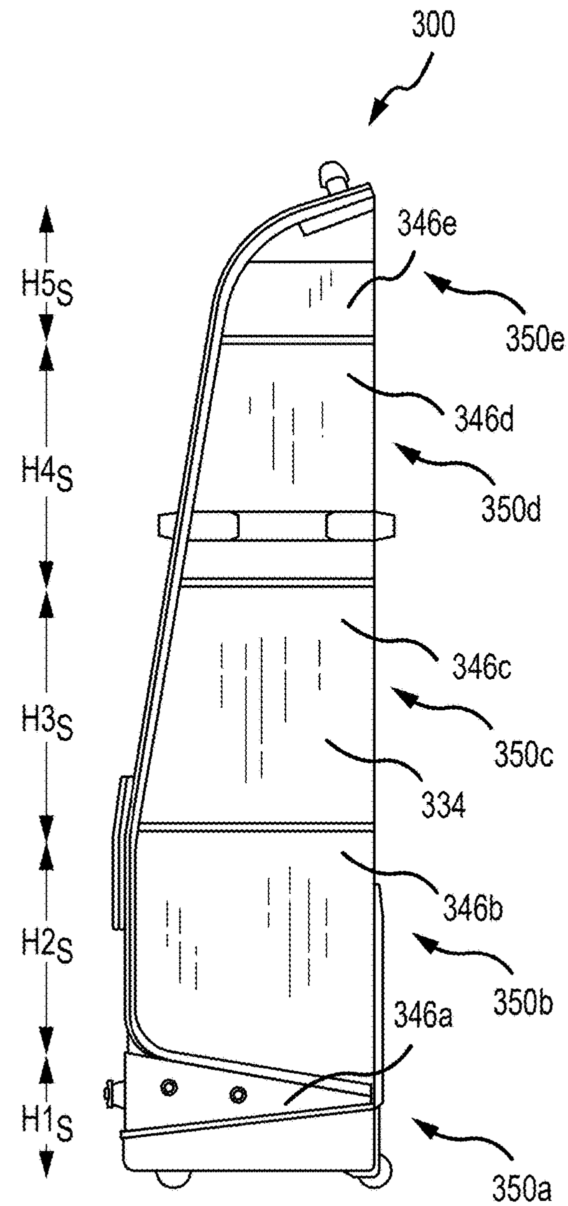

FIG.28B

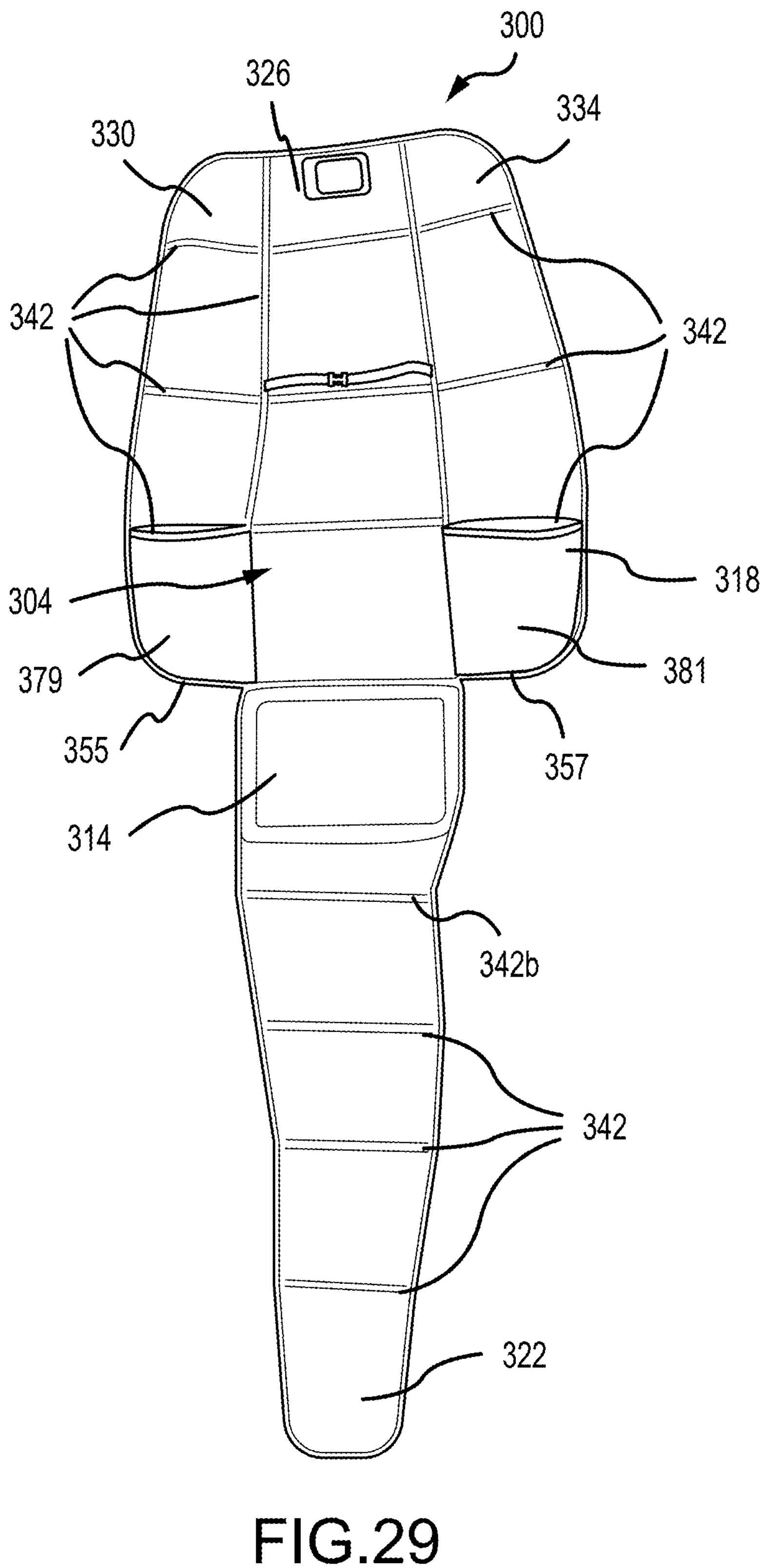

FIG.29

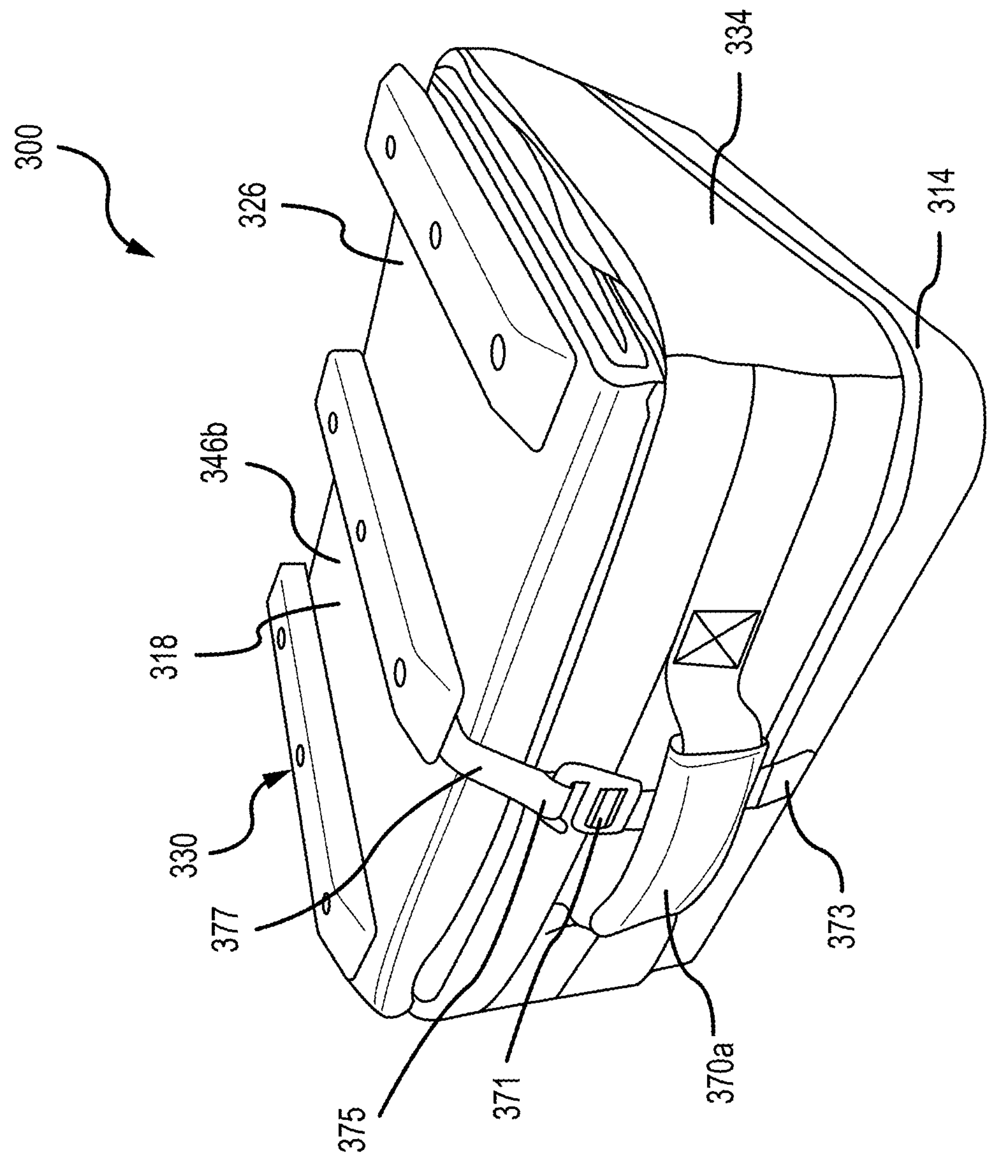

FIG.30

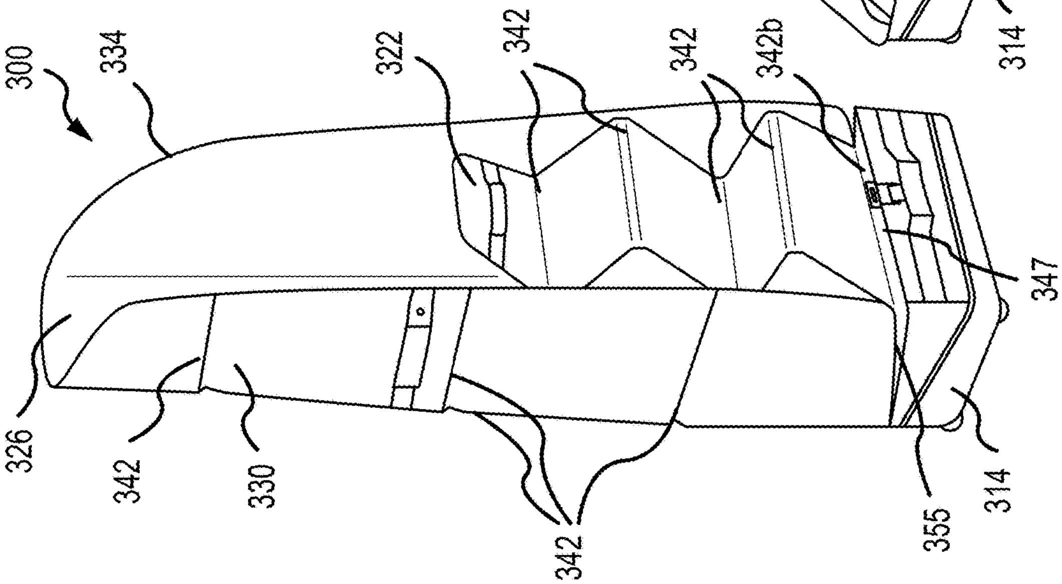

FIG.31A

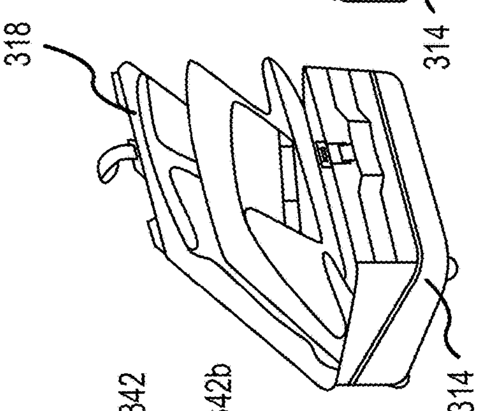

FIG.31B

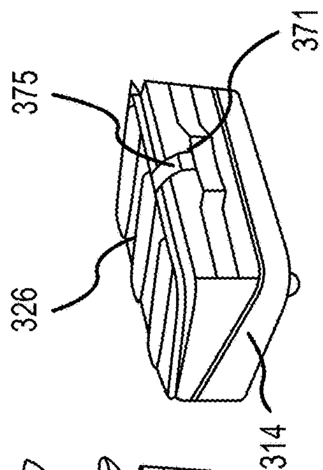

FIG.31C

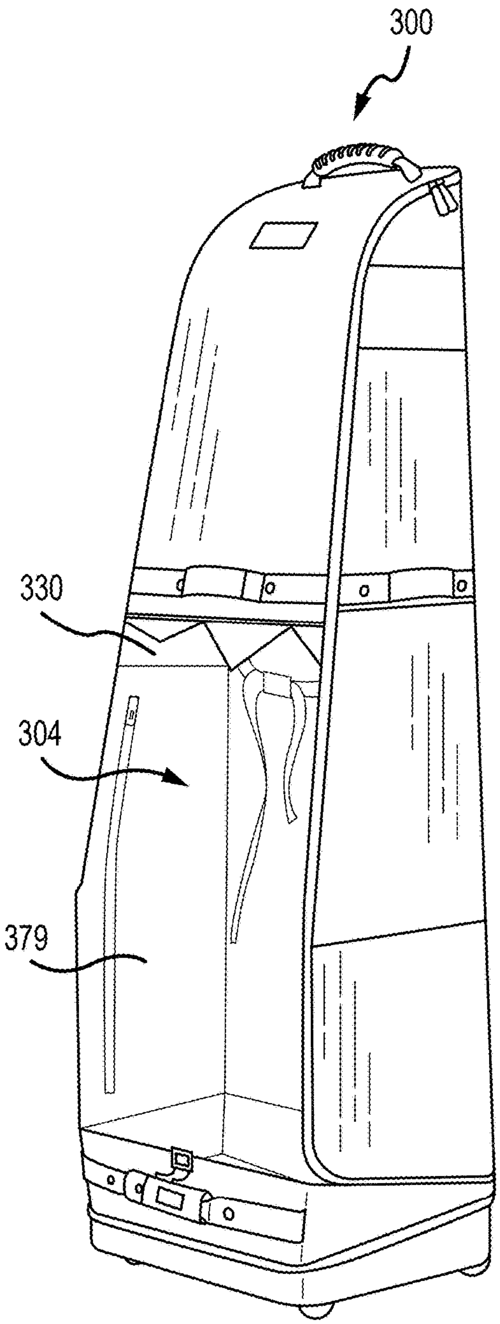

FIG.32

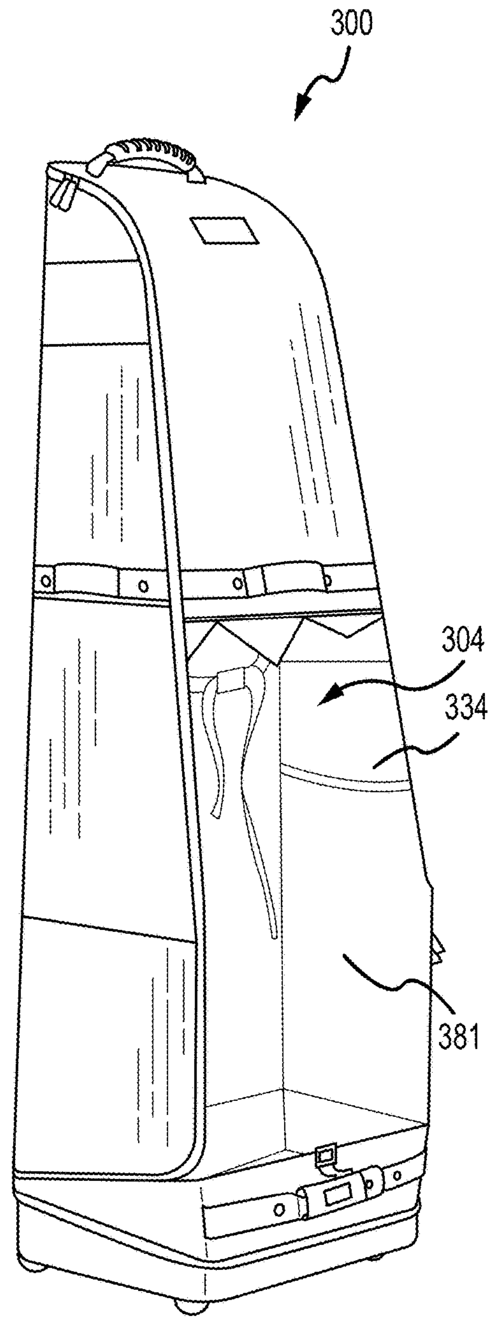

FIG.33

ROLLING COLLAPSIBLE TRAVEL LUGGAGE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a continuation of U.S. Non-Provisional application Ser. No. 18/066,098, filed on Dec. 14, 2022, which is a continuation in part of U.S. Non-Provisional application Ser. No. 17/835,842, filed on Jun. 8, 2022, which is a continuation of U.S. Non-Provisional application Ser. No. 16/439,542, filed on Jun. 12, 2019, which is a continuation in part of U.S. Non-Provisional application Ser. No. 16/163,371, filed on Oct. 17, 2018, which is a continuation of U.S. Non-Provisional application Ser. No. 15/000,280, filed on Jan. 19, 2016, which claims priority to U.S. Provisional Patent Application No. 62/105,636, filed on Jan. 20, 2015, and U.S. Provisional Patent Application No. 62/189,598, filed on Jul. 7, 2015. U.S. Non-Provisional application Ser. No. 16/439,542 further claims priority to U.S. Provisional Patent Application No. 62/684,133, filed on Jun. 12, 2018, all of which are incorporated by reference herein in their entirety.

FIELD OF THE INVENTION

[0002] The present disclosure relates to rolling collapsible travel luggage, and more specifically to wheeled travel luggage that is easier to transport when in use, which is collapsible to provide a smaller storage footprint when not in use, and that has an expanded access opening to more easily place and position contents into the luggage.

BACKGROUND

[0003] Rolling travel luggage is generally known in the art. However, known rolling travel luggage has certain limitations. For example, existing luggage typically includes three or more swivel caster wheels that allow the luggage to roll in any direction while in an upright position. While convenient, the luggage is susceptible to movement in unintended directions as the only wheels that engage the ground or floor are the swivel caster wheels, with nothing to stop unintended rotation or rolling of these wheels.

[0004] As another limitation, existing luggage typically defines an internal chamber by rigid or relatively inflexible side, back and/or front portions. For example, the front side may open away from the remaining rigid sides to provide an access opening to the internal chamber. While the rigid sides provide protection for the contents inside, they inhibit insertion of a large or otherwise bulky object, such as a golf bag containing a set of golf clubs. To place the large or bulky object into the internal chamber, a user must feed the object into the access opening at an oblique or other awkward angle to the luggage. And during insertion, the user may be required to constantly change the angle between the object and luggage to avoid contact with the rigid sides until the object is received in the internal chamber. The constant angle change can be cumbersome, difficult, and awkward for a user.

[0005] As yet another limitation, some types of existing luggage include side wheels and a handle provided at a top portion of the luggage. A user grasps the handle, tilts the luggage to engage the side wheels with the ground, and is free to roll the luggage in the tilted position. The majority of the luggage load, however, is transferred to the user through the top handle, leading to strain on the arm, wrist, and/or forearm of the user.

SUMMARY OF THE INVENTION

[0006] A rolling luggage bag includes a cover coupled to a base, the base including a first side opposite a second side and a bottom face extending there between, a first wheel and a second wheel coupled to the base, the first and second wheels configured to rotate about an axis of rotation and separated by a first distance along the axis of rotation, at least a portion of each of the first and second wheels projecting from the first side and from the bottom face, and a third wheel and a fourth wheel are eacoupled to the bottom face, the third and fourth wheels configured to independently swivel about a respective swivel axis and separated by a second distance extending between the swivel axes. When the rolling luggage is in an upright position, the first, second, third, and fourth wheels all contact a surface the luggage bag stands on.

[0007] In other embodiments, the collapsible luggage bag can include a back portion connected to a first side portion and a second side portion, a front flap removably connected to the first side portion, the second side portion, and the back portion by a connection member, and a base connected to the back portion, the front flap, and optionally removable from the first side portion, and the second side portion. The first and second side portions pivot away from each other about respective folds between the respective side portion and the back portion when the front flap is removed.

[0008] Other features and aspects will become apparent by consideration of the following detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a perspective view of a rolling collapsible travel luggage bag in an upright position.

[0010] FIG. **2** is a perspective view from the top of the rolling collapsible travel luggage bag of FIG. **1** taken along line **2-2** of FIG. **1**.

[0011] FIG. **3** is a perspective view from a first side of the rolling collapsible travel luggage bag of FIG. **1** taken along line **3-3** of FIG. **2**.

[0012] FIG. **4**A is a perspective view from the rear of the rolling collapsible travel luggage bag of FIG. **1** taken along line **4-4** of FIG. **3**, illustrating a first position of a rolling handle.

[0013] FIG. **4**B is a perspective view from the rear of the rolling collapsible travel luggage bag of FIG. **1** taken along line **4-4** of FIG. **3**, illustrating a second position of the rolling handle.

[0014] FIG. **5** is a perspective view from a second side of the rolling collapsible travel luggage bag of FIG. **1** taken along line **5-5** of FIG. **4**A.

[0015] FIG. **6** is a perspective view from the rear of the rolling collapsible travel luggage bag of FIG. **1** taken along line **6-6** of FIG. **5**.

[0016] FIG. **6**A is a perspective view of an alternative embodiment of the rails for use with the rolling collapsible travel luggage bag illustrated in FIGS. **5** and **6** and shown in an upright position.

[0017] FIG. **6**B is a side view of the rolling collapsible travel luggage bag of FIG. **6**A.

[0018] FIG. **6**C is a perspective view of the rolling collapsible travel luggage bag of FIG. **6**A shown in a collapsed, folded position.

[0019] FIG. **6**D is a side view of the rolling collapsible travel luggage bag of FIG. **6**C.

[0020] FIG. **7**A is an elevation view of the second side portion of the rolling collapsible travel luggage bag of FIG. **1**, with the outer shell removed to illustrate the respective panels.

[0021] FIG. **7**B is an elevation view of the back portion of the rolling collapsible travel luggage bag of FIG. **1**, with the outer shell removed to illustrate the respective panels.

[0022] FIG. **7**C is an elevation view of the first side portion of the rolling collapsible travel luggage bag of FIG. **1**, with the outer shell removed to illustrate the respective panels.

[0023] FIG. **7**D is an elevation view of the front flap of the rolling collapsible travel luggage bag of FIG. **1**, with the outer shell removed to illustrate the respective panels.

[0024] FIG. **8** is a perspective view of the rolling collapsible travel luggage bag of FIG. **1** in the upright position with the front flap partially disengaged to provide access to an interior chamber.

[0025] FIG. **9** is a partial view of the rolling collapsible travel luggage bag of FIG. **1** illustrating a butterfly opening providing access to the interior chamber without obstruction from a portion of a cover, and with a portion of the front flap shown.

[0026] FIG. **10** is a bottom plan view of the rolling collapsible travel luggage bag of FIG. **1**, illustrating an external bottom surface of the luggage bag base.

[0027] FIG. **11** is a top plan view of an internal bottom surface of the base of FIG. **10**.

[0028] FIG. **12** is a front side view of the base taken along line **12-12** of FIG. **10**.

[0029] FIG. **13** is a back side view of the base taken along line **13-13** of FIG. **10**.

[0030] FIG. **14** is a first side view of the base taken along line **14-14** of FIG. **10**.

[0031] FIG. **15** is a second side view of the base taken along line **15-15** of FIG. **10**.

[0032] FIG. **16** is a perspective view of the rolling collapsible travel luggage bag of FIG. **1** in the upright position with the front flap disengaged from a first side portion, a second side portion, and a back portion.

[0033] FIG. **17** is a perspective view of the rolling collapsible travel luggage bag of FIG. **1** in a partially collapsed position with a portion of the cover received by a base.

[0034] FIG. **18**A is a side elevation view of a storage bag containing the luggage bag of FIG. **1** in a collapsed position

[0035] FIG. **18**B is another side elevation view of the storage bag of FIG. **18**A, with a portion of the storage bag shown in broken lines to illustrate the luggage bag of FIG. **1** in the collapsed position in the bag.

[0036] FIG. **19** is a schematic view of the rolling collapsible travel luggage bag of FIG. **1** in a tilted position, illustrating certain forces on the luggage.

[0037] FIG. **20** is a front view of an embodiment of a collapsible travel luggage bag in an upright position.

[0038] FIG. **21** is a rear view of the collapsible travel luggage bag of FIG. **20**, in an upright position.

[0039] FIG. **22** is a first (right) side view of the collapsible travel luggage bag of FIG. **20**, in an upright position.

[0040] FIG. **23** is a second (right) side view of the collapsible travel luggage bag of FIG. **20**, in an upright position.

[0041] FIG. **24** is a perspective view an internal cover of a collapsible travel luggage bag.

[0042] FIG. **25** is a front, x-ray view of the internal cover of FIG. **24**, installed into a collapsible travel luggage bag.

[0043] FIG. **26** is a front perspective view of an embodiment of a storage bag.

[0044] FIG. **27**A is a front perspective view of an embodiment of a collapsible travel luggage bag in an upright position.

[0045] FIG. **27**B is a rear perspective view of the collapsible travel luggage bag of FIG. **27**A, in an upright position.

[0046] FIG. **28**A is a front view of the collapsible travel luggage bag of FIG. **27**A, in an upright position.

[0047] FIG. **28**B is a side view of the collapsible travel luggage bag of FIG. **27**A, in an upright position.

[0048] FIG. **29** is a perspective view of the collapsible travel luggage bag of FIG. **27**A with the front flap disengaged to provide access to an interior chamber.

[0049] FIG. **30** is a perspective view of the travel luggage bag of FIG. **27**A in a collapsed position with the cover received by a base.

[0050] FIG. **31**A is a perspective view of the travel luggage bag of FIG. **27**A in the upright position with the front flap disengaged from a first side portion, a second side portion, and a back portion, and the first side portion and the second side portion disengaged from a base.

[0051] FIG. **31**B is a perspective view of the travel luggage bag of FIG. **27**A in a partially collapsed position with the cover partially received by a base.

[0052] FIG. **31**C is a perspective view of the travel luggage bag of FIG. **27**A in a collapsed position with the cover received by a base.

[0053] FIG. **32** is a partial view of the collapsible travel luggage bag of FIG. **27**A illustrating an inner pocket.

[0054] FIG. **33** is a partial view of the collapsible travel luggage bag of FIG. **27**A illustrating an inner pocket.

[0055] Before any embodiments of the disclosure are explained in detail, it should be understood that the disclosure is not limited in its application to the details or construction and the arrangement of components as set forth in the following description or as illustrated in the drawings. The disclosure is capable of supporting other embodiments and of being practiced or of being carried out in various ways. It should be understood that the description of specific embodiments is not intended to limit the disclosure from covering all modifications, equivalents and alternatives falling within the spirit and scope of the disclosure. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

[0056] For ease of discussion and understanding, and for purposes of description only, the following detailed description illustrates a rolling luggage bag **10** as an elongated luggage bag suitable for transporting large items, such as a golf bag, golf clubs, and one or more golf accessories. The golf bag is of a size suitable to carry a plurality of full length golf clubs, for example a set of golf clubs that includes a combination of one or more of a driver, a wood, a hybrid, an iron, a wedge, and/or a putter. A full length golf club is not collapsible, and has a length of approximately 32 inches to approximately 49 inches, depending on the club. An example of a large golf accessory includes a full length golf umbrella, which has a length of approximately 36 inches to approximately 48 inches, and opens to an arc or canopy size of approximately 50 inches to 68 or more inches. It should be appreciated that the elongated luggage bag is provided for purposes of illustration and aspects of the luggage bag **10** disclosed herein may be incorporated into luggage of any size, shape, or orientation.

[0057] FIG. **1** illustrates a rolling luggage bag **10**. The luggage bag **10** includes a base **14** connected to a cover **18**. As illustrated in FIGS. **2-6**, the cover **18** includes a front portion **22** (or front flap), a back portion **26** or (back flap), a first side portion **30** (flap **30**), and a second side portion **34** (or flap). The front portion **22** is oriented opposite the back portion **26**, and the first side portion **30** is oriented opposite the second side portion **34**. In addition, the back portion **26** is connected or otherwise integrally formed as one piece with the first and second side portions **30**, **34**. The front portion **22** may have a selectively removable connection to the back, first side, and second side portions **26**, **30**, **34** to open or provide access to the interior chamber of the luggage bag **10**. The selective removable connection between the front portion **22** and back, first side, and second side portions **26**, **30**, **34** may provide different degrees or amounts of access to the interior chamber of the luggage bag **10**. Stated otherwise, the removable connection may provide an opening or partial opening to the interior chamber. For example, the selectively removable connection may be formed by a zipper having a pair of sliders **100***a*, **100***b* that meet in a closed position at the top or end of the luggage bag **10** opposite the base **14** (see FIGS. **4**A-B). The sliders **100***a*, **100***b* may be moved along the zipper in opposite directions along a portion of the zipper to provide partial access to the interior chamber, such as a first standing access position shown in FIG. **8**, or entirely along the zipper to provide maximum access to the interior chamber, such as a second access position shown in FIG. **9**.

[0058] The front, back, first side, and second side portions **22**, **26**, **30**, **34** are each formed of a plurality of panels interconnected by an outer shell **38** (see FIG. **16**). In most embodiments, the outer shell **38** is formed of a cloth or other fabric. For example, the fabric may be polyester, nylon, canvas, denim, or any other fabric material suitable for use in a luggage type application. In each of the portions **22**, **26**, **30**, **34**, the outer shell **38** includes stitching to seal the outer shell **38** and to define a plurality of pockets, each of which receives a respective panel. Between adjacent pockets in each portion **22**, **26**, **30**, **34** are folds **42** (see FIGS. **1-9**, and **16**). To form the folds **42**, stitching is provided at desired fold locations on the outer shell **38** of each portion **22**, **26**, **30**, **34**. In other embodiments, the folds **42** may be formed in any other suitable or desired manner to facilitate folding of each portion **22**, **26**, **30**, **34**. The folds **42** are not only provided between adjacent or consecutive panels in each portion **22**, **26**, **30**, **34**, but are generally provided between adjacent or consecutive panels between portions **26**, **30**, **34**.

[0059] Referring to FIGS. **7**A-7D, each portion **22**, **26**, **30**, **34** is shown with the outer shell **38** removed and illustrating the plurality of panels **46**. In most embodiments, each panel **46** is formed of a polyethylene board material $[(C_2H_2)_nH_2]$, a foam material, a cloth material, or a combination thereof. The polyethylene board material may be any industry standard grade, including, but not limited to, ultra-high molecular weight polyethylene (UHMWPE), ultra-low molecular weight polyethylene (ULMWPE), high molecular weight polyethylene (HMWPE), high-density polyethylene (HDPE), high-density cross-linked polyethylene (HDXLPE), cross-linked polyethylene (PEX), medium-density polyethylene (MDPE), linear low-density polyethylene (LLDPE), low-density polyethylene (LDPE), very-low density polyethylene (VLDPE), and chlorinated polyethylene (CPE). In other embodiments, the panels **46** may be formed of polyurethane, acrylonitrile butadiene styrene, combinations thereof, or any other suitable material.

[0060] The panels **46** of each portion **22**, **26**, **30**, **34** are arranged in panel zones, panel rings, or bands that are horizontally stacked or arranged in a direction away from the base **14**. The panel zones **50** generally extend around a circumference of the luggage bag **10**. Depending on the panel zone **50** and location along portions **22**, **26**, **30**, **34**, any panel **46** may be formed of a different material (e.g., a foam material, a board material, or a combination of foam and board material), may have a different panel thickness, and/or a different panel stiffness or rigidity.

[0061] A first or bottom panel zone **50***a* is located along a base of each portion **22**, **26**, **30**, **34** and has a first panel height or length H_1, measured vertically (orthogonal to a ground plane) when the luggage bag **10** is in an upright position. In some embodiments, the first panel height H_1 is less than approximately 15 cm. The first panel height H_1 can be approximately 5.0 cm, 5.5 cm, 6.0 cm, 6.5 cm, 7.0 cm, 7.5 cm, 8.0 cm, 8.5 cm, 9.0 cm, 9.5 cm, 10.0 cm, 10.5 cm, 11.0 cm, 11.5 cm, 12 cm, 12.5 cm, 13.0 cm, 13.5 cm, 14.0 cm, 14.5 cm, or 15.0 cm. In one exemplary embodiment, first panel height H_1 is approximately 6.5 cm. In other embodiments, the first panel height H_1 can be less than 6.5 cm, 6.4 cm, 6.3 cm, 6.2 cm, 6.1 cm, or 6.0 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the first panel height H_1 can be greater than 6.5 cm, 6.6 cm, 6.7 cm, 6.8 cm, 6.9 cm, or 7.0 cm based on the size, shape, or orientation of the luggage bag **10**. The first panel zone **50***a* includes panels **46***a* formed of a stiff board material having a thickness of approximately 1.75 mm. In some embodiments, the panels **46***a* of the first panel zone **50***a* are rectangular in shape, however in other embodiments may be any suitable or desired shape or combination of shapes. For example, the panels **46***a* of the first panel zone **50***a* may be sloped or have a narrowing height H_1 (see FIG. **3**) to conform to the base **14**. The panels **46***a* of the first panel zone **50***a* engage or otherwise connect to the base **14** (see FIG. **3**).

[0062] Positioned adjacent the first panel zone **50***a* in a direction away from base **14** is a second panel zone **50***b* along each portion **22**, **26**, **30**, **34**. The second panel zone **50***b* has a second panel height or length H_2, measured vertically (orthogonal to a ground plane) when the luggage bag **10** is in an upright position. In some embodiments, the height H_2 can be less than 50 cm. In some embodiments, the second panel height or length H_2 can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, 48 cm, 49 cm, or 50 cm. In one exemplary embodiment, the second panel height H_2 is approximately 30 cm. In other embodiments, the second panel height H_2 can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the second panel height H_2 can be greater than 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**. The front portion **22** of the second panel zone **50***b* includes a panel **46***b*. The back, first side, and second side portions **26**, **30**, **34** of the second panel zone **50***b* include panels **46***b* formed of a relatively stiff board material combined with foam. In many embodiments, the stiff board material has a thickness of approximately 2.5 mm. The positioning of the panels **46***b* near or approximate the base **14** provides structural support for the back, first side, and second side portions **26**, **30**, **34** while the luggage bag **10** is in the upright position, as illustrated in FIG. **8**.

[0063] In some embodiments, the panels of the second panel zone **50***b* in the front and back portions **22**, **26** have a generally rectangular shape while the panels in the first side and second side portions **30**, **34** have a generally square shape; however, this square shape is defined by two separate triangular panels having an angled fold **42***a* there between, which facilitates collapsibility of the cover **18** (discussed in more detail below). In some embodiments, the panels in the first side and second side portions **30**, **34** have a generally trapezoidal shape. In other embodiments, the panels of the second panel zone **50***b* may be any suitable or desired shape or combination of shapes.

[0064] Adjacent the second panel zone **50***b* in a direction away from base **14** is a third panel zone **50***c* along each portion **22**, **26**, **30**, **34**. The third panel zone **50***c* has a third panel height or length H_3, measured vertically (orthogonal to a ground plane) when the luggage bag **10** is in an upright position. In some embodiments, the third panel height or length H_3 can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, or 40 cm. In one exemplary embodiment, the third panel height H_3 is approximately 32 cm. In other embodiments, the third panel height H_3 can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the third panel height H_3 can be greater than 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**. The front portion **22** and side portions **30**, **34** of the third panel zone **50***c* each include a panel **46***c* formed of a relatively stiff board material combined with foam. In many embodiments, the stiff board material has a thickness of approximately 1.0 mm. The back portion **26** of the third panel zone **50***c* includes a panel **46***a* as previously described. In some embodiments, the panels of the third panel zone **50***c* in the front, back, first side, and second side portions **22**, **26**, **30**, **34** have a generally trapezoidal shape. In other embodiments, the panels of the third panel zone **50***c* in the front, back, first side, and second side portions **22**, **26**, **30**, **34** have a generally rectangular shape. In other embodiments, the panels of the third panel zone **50***c* may be any suitable or desired shape or combination of shapes.

[0065] A fourth panel zone **50***d* is adjacent the third panel zone **50***c* in a direction away from base **14** along each portion **22**, **26**, **30**, **34**, and has a fourth panel height or length H_4, measured vertically (orthogonal to a ground plane) when the luggage bag **10** is in an upright position. In some embodiments, the fourth panel height or length H_4 can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, or 40 cm. In one exemplary embodiment, the fourth panel height H_4 is approximately 32 cm. In other embodiments, the fourth panel height H_4 can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the fourth panel height H_4 can be greater than 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**. In many embodiments, the fourth panel zone **50***d* is substantially the same as the third panel zone **50***c* with regard to panel type, positioning, and shape.

[0066] At the top of each portion **22**, **26**, **30**, **34** furthest from the base **14** is a fifth panel zone or top panel zone **50***c*. The back, first side, and second side portions **26**, **30**, **34** of the top panel zone **50***c* have a fifth panel height or length H_5, measured vertically (orthogonal to a ground plane) when the luggage bag **10** is in an upright position. In some embodiments, the fifth panel height or length H_5 can be 10 cm, 11 cm, 12 cm, 13 cm, 14 cm, 15 cm, 16 cm, 17 cm, 18 cm, 19 cm, 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, or 30 cm. In one exemplary embodiment, the fifth panel height H_5 is approximately 20 cm. In other embodiments, the fifth panel height H_5 can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the fifth panel height H_5 can be greater than 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In some embodiments, the back portion **26** includes a panel in the top panel zone **50***c* generally trapezoidal in shape and having an end edge **54**. The first and second side portions **30**, **34** each include a panel in the top panel zone **50***c* that has a respective arcuate edge **58***a*, **58***b*. In such an embodiment, the arcuate edges **58***a*, **58***b* lead to the end edge **54** of the back portion **26** (see FIGS. **8** and **9**).

[0067] The front portion **22** of the top panel zone **50***e* has a sixth panel height or length H_6, measured vertically (orthogonal to a ground plane) when the luggage bag **10** is in an upright position. In many embodiments, sixth panel height H_6 is generally greater than the fifth panel height H_5. In some embodiments, the sixth panel height or length H_6 can be 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, or 48 cm. In one exemplary embodiment, the sixth panel height H_6 is approximately 38.5 cm. In other embodiments, the sixth panel height H_6 can be less than 40 cm, 39 cm, 38 cm, 37 cm, 36 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the sixth panel height H_6 can be greater than 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**. In some embodiments, the panel within zone **50***c* of the front portion **22** includes a parabolic edge **62** that removably connects to the arcuate edges **58***a*, **58***b* and to the end edge **54**, and the additional panel height H_6 of the front portion **22** defines or forms a curved or an arcuate face **66** (see FIGS. **2-3**) that extends over or overlaps a portion of a base footprint defined by a periphery of the base **14**, as best illustrated in FIG. **2**. In such an embodiment, the arcuate face **66** removably connects to the back, first side, and second side portions **26**, **30**, **34**.

[0068] In some embodiments, the panel heights H_1, H_2, H_3, and H_4 vary across the front, back, first side, and second side sections. In some embodiments, the front portion **22** is slightly angled after assembly and the back portion **26** is approximately vertical, with respect to the ground plane when the luggage bag **10** in an upright position. Due to the angled orientation of the front portion **22**, when the front section is laid out flat, as illustrated in FIG. 7D, the panel heights of the front portion **22** can be slightly greater than the panel heights of the back **26**, the first side portion **30**, and the second side **34** portion (height difference is not visible in the illustration). Similarly, in some embodiments, the panel height of the first side portion **30** and the second side portion **34** can be slightly greater than the panel heights of the back portion **26**. In some embodiments, the differences in panel section heights can be minimal, varying by a length between 0 cm and 0.2 cm. However, in other embodiments, the fourth panel height H_4 is greater in the front portion **22** than in the back portion **26** by a length between 1.5 cm and 3.5 cm. Embodiments, similar to the embodiment of FIGS. **22** and **23**, have a fourth panel **50***d* that is angled in the front portion **22**. Thus, these embodiments comprise a greater difference in the fourth panel height H_4 between sections.

[0069] The front portion **22** can have a width measured parallel to the folds **42** from a point where the front portion **22** intersects the first side portion **30** to a point where the front portion **22** intersects the second side portion **34**. The back portion **26** can have a width measured parallel to the folds **42** from a point where the back portion **26** intersects the first side portion **30** to a point where the back portion **26** intersects the second side portion **34**. In many embodiments, the width of the front portion **22** is the same as or similar to the width of the back portion **26** at any given height.

[0070] A first width W_1 of the back portion **26** is measured at the fold line between the first panel zone **50***a* and the second panel zone **50***b*. The first width W_1 can range between 35 cm and 50 cm. In some embodiments, the first width W_1 can be 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, 48 cm, 49 cm, or 50 cm. In the illustrated embodiment, the first width W_1 is approximately 43 cm. In many embodiments, a first width of the front portion **22** is the same as or similar to the first width W_1 of the back portion **26**.

[0071] A second width W_2 of the back portion **26** is measured at the fold line between the second panel zone **50***b* and the third panel zone **50***c*. The second width W_2 can range between 35 cm and 50 cm or between 35 cm and 40 cm. In some embodiments, the second width W_2 can be 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, 48 cm, 49 cm, or 50 cm. In the illustrated embodiment, the second width W_2 is approximately 39 cm. In many embodiments, a second width of the front portion **22** is the same as or similar to the second width W_2 of the back portion **26**.

[0072] A third width W_3 of the back portion **26** is measured at the fold line between the third panel zone **50***c* and the fourth panel zone **50***d*. The third width W_3 can range between 25 cm and 45 cm or between 30 cm and 40 cm. In some embodiments, the third width W_3 can be 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, 48 cm, 49 cm, or 50 cm. In the illustrated embodiment, the third width W_3 is approximately 39 cm. In many embodiments, a third width of the front portion **22** is the same as or similar to the third width W_3 of the back portion **26**.

[0073] A fourth width W_4 of the back portion **26** is measured at the fold line between the fourth panel zone **50***d* and the fifth panel zone **50***c*. The fourth width W_4 can range between 20 cm and 45 cm or between 20 cm and 30 cm. In some embodiments, the fourth width W_4 can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, or 45 cm. In the embodiment illustrated in FIGS. **20-23**, the fourth width W_4 is approximately 27 cm. In many embodiments, a fourth width of the front portion **22** is the same as or similar to the fourth width W_4 of the back portion **26**.

[0074] The first side portion **30** can have a depth measured parallel to the folds **42** from a point where the first side portion **30** intersects the back portion **26** to a point where the first side portion **30** intersects the front portion **22**. The second side portion **34** can have a depth measured parallel to the folds **42** from a point where the second side portion **34** intersects the back portion **26** to a point where the second side portion **34** intersects the front portion **22**. At any given height, the depth of the first side portion **30** is the same as the depth of the second side portion. The depths of each portion are carefully selected to allow various golf bags to be secured within the interior chamber **102**.

[0075] A first depth D**1** of the second side portion **34** is measured at the fold line between the first panel zone **50***a* and the second panel zone **50***b*. In many embodiments, the first depth D_1 can range between 20 cm and 40 cm, or between 25 cm and 35 cm. In some embodiments, the first depth D_1 can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, or 40 cm. In one exemplary embodiment, the first depth D_1 is approximately 30 cm. In other embodiments, the first depth D_1 can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the first depth D_1 can be greater than 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**. In many embodiments, a first depth of the first side portion **30** is the same as the first depth D_1 of the second side portion **34**.

[0076] A second depth D_2 of the second side portion **34** is measured at the fold line between the second panel zone **50***b* and the third panel zone **50***c*. In many embodiments, the second depth D_2 can range between 20 cm and 35 cm, or between 25 cm and 35 cm. In some embodiments, the second depth D_2 can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm. In one exemplary embodiment, the second depth D_2 is approximately 29 cm. In other embodiments, the second depth D_2 can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the second depth D_2 can be greater than 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, or 30 cm based on the size, shape, or orientation of the luggage bag **10**. In many embodiments, a second depth of the first side portion **30** is the same as the second depth D_2 of the second side portion **34**.

[0077] A third depth D_3 of the second side portion **34** is measured at the fold line between the third panel zone **50***c* and the fourth panel zone **50***d*. In many embodiments, the third depth D_3 can range between 20 cm and 35 cm, or between 25 cm and 30 cm. In some embodiments, the third depth D_3 can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm. In one exemplary embodiment, the third depth D_3 is approximately 26 cm. In other embodiments, the third depth D_3 can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the third depth D_3 can be greater than 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, or 30 cm based on the size, shape, or orientation of the luggage bag **10**. In many embodiments, the third depth of the first side portion **30** is the same as the third depth D_3 of the second side portion **34**.

[0078] A fourth depth D_4 of the second side portion **34** is measured at the fold line between the fourth panel zone **50***d* and the fifth panel zone **50***c*. In many embodiments, the fourth depth D_4 can range between 10 cm and 35 cm, or between 12 cm and 20 cm. In some embodiments, the fourth depth D_4 can be 10 cm, 11 cm, 12 cm, 13 cm, 14 cm, 15 cm, 16 cm, 17 cm, 18 cm, 19 cm, 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm. In one exemplary embodiment, the fourth depth D_4 is approximately 16 cm. In other embodiments, the fourth depth D_4 can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the fourth depth D_4 can be greater than 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, or 30 cm based on the size, shape, or orientation of the luggage bag **10**. In many embodiments, the fourth depth of the first side portion **30** is the same as the fourth depth D_4 of the second side portion **34**.

[0079] The back portion **26** includes in the top panel zone **50***e* a panel **46***c* formed of a relatively stiff board material combined with foam. The stiff board material has a thickness of approximately 1.0 mm. The front, first side, and second side portions **22**, **30**, **34** include in the top panel zone **50***c* a panel **46***d* formed of foam and that does not include a board material. Forming the panels **46***d* from foam provides flexibility to the front, first side, and second side portions **22**, **30**, **34** to facilitate formation of the arcuate face **66** (see FIGS. **2-3**) of the front portion **22** in the top panel zone **50***c*.

[0080] It should be appreciated that a greater thickness of the board material leads to a more rigid or a greater stiffness panel. For example, the panels in the second panel zone **50***b* are more rigid and/or have a greater stiffness than the panels in the top panel zone **50***c*. Generally, the overall stiffness or rigidity of the panels decreases from the base **14** upward toward the arcuate face **66**.

[0081] The cover **18** can comprise between 15 to 30 total panels **46**. In some embodiments, the cover **18** comprises between 15 to 20, 17 to 25, 20 to 25, or 25 to 30 total panels **46**. For example, the cover **18** illustrated in FIG. **3** comprises twenty-two total panels **46**. In some embodiments, the cover **18** comprises at least 20 panels, 21 panels, 22 panels, 23 panels, 24 panels, or 25 panels. In other embodiments, the cover **18** comprises no more than 25 panels, 24 panels, 23 panels, 22 panels, 21 panels, or 20 panels. The total number of panels may be based on the size, shape, and/or collapsibility of the luggage bag **10**.

[0082] In one exemplary embodiment, the luggage bag **10** has a height or length, defined by the sum of heights H_1 to H_5, of approximately 120.5 cm (or approximately 47.5 inches). In other embodiments, the luggage bag **10** may have a height or length in a range of approximately 110 cm to approximately 140 or more cm. Stated another way, the luggage bag **10** may have a height or length suitable to receive a golf bag and/or one or more full length golf clubs.

[0083] Referring back to FIGS. **1-6**, the luggage bag **10** includes a plurality of handles **70** to assist with lifting and otherwise carrying the luggage bag **10**. As illustrated in FIGS. **1** and **3**, a first handle **70***a* is connected to the front portion **22** in the first panel zone **50***a* proximate or near the base **14** to provide a user a location to grasp near the base **14**. Referring to FIGS. **1-6**, a plurality of second handles **70***b*, **70***c*, **70***d*, **70***e* are respectively connected to the front portion **22**, back portion **26**, first side portion **30**, and second side portion **34**. The second handles **70***b*, **70***c*, **70***d*, **70***e* are illustrated in the same horizontal plane around the luggage bag **10**, approximately 70 cm to 90 cm from the base **14** (or surface on which the base **14** is positioned). In other embodiments, the second handles **70***b*, **70***c*, **70***d*, **70***c* may be offset, staggered, or positioned on panels in other panel zones **50**, and at various distances from the base **14** (or surface on which the base **14** is positioned). In addition, fewer or more than four second handles **70***b*, **70***c*, **70***d*, **70***e* may be connected to luggage bag **10**. The first handle **70***a* and second handles **70***b*, **70***c*, **70***d*, **70***c* may be any suitable handle for use with luggage. In the illustrated embodiments, the handles **70***a*, **70***b*, **70***c*, **70***d*, **70***e* are formed of a durable fabric, and include a handle wrap. In other embodiments, the handles **70***a*, **70***b*, **70***c*, **70***d*, **70***c* may be formed of any suitable materials.

[0084] Referring to FIGS. **4**A, **5**, and **6**, the luggage bag **10** includes a rolling handle **74** provided on a panel or top back panel **78** on the back side **26** within the top panel zone **50***c*. In the illustrated embodiment, the rolling handle **74** projects away from the back side **26**, and is located adjacent or towards the end edge **54** (see FIG. **8**) of the panel **78**. In other embodiments, the handle **74** may be located at any desired location along the panel **78**. The rolling handle **74** is shown as reinforced with a rubber handle wrap but can be any other suitable handle for use with luggage bag **10**. In yet other embodiments, and as illustrated in FIG. **4**B, the rolling handle **74** may instead be positioned on the arcuate face **66**. Alternatively, the luggage bag **10** may include two rolling handles **74**, with one positioned on the panel **78** (for example the position illustrated in FIG. **4**A) and the other on the arcuate face **66** (for example the position illustrated in FIG. **4**B).

[0085] Referring now to FIGS. **5** and **6**, the luggage bag **10** includes a bumper in the form of rails **82** positioned on a panel or bottom back panel **86** within the second panel zone **50***b*, adjacent but separate from the base **14**. The rails **82** extend upward, away from the base **14** and provide protection against luggage damage when the luggage bag **10** is pulled over a curb or other uneven surface. In the illustrated embodiment, the rails **82** are arranged on the single panel **86** and do not extend to any adjacent panel or to the base **14**, thereby facilitating collapsibility of the luggage bag **10** (further detailed below). In one or more examples of embodiments, the rails **82** are approximately 20 cm to 30 cm long with a width of approximately 2 cm to 5 cm. In other embodiments the rails **82** may be any length or width suitable for providing protection against damage to the luggage bag **10** while facilitating collapsibility of the luggage bag **10**. In the illustrated embodiment, the luggage bag **10** includes three total rails **82**, with outermost first and second rails **82** in respective vertical alignment with wheels **136***a*, **136***b*. A third middle rail **82** is approximately equidistant from the outermost rails **82**. In other embodiments, any suitable number of rails **82** may be implemented, for example two rails **82** or four or more rails **82**. In yet other embodiments, additional bumpers, or rails **82** may be positioned on other panels, including in the first or bottom panel zone **50***a* or on the base **14**.

[0086] FIGS. **6**A-**6**D illustrate an alternative embodiment of the luggage bag **10** having a bumper or bumper assembly **180** that extends from the base **14** along a portion of the cover **18**. The bumper **180** includes rails **182** that removably engage a corresponding rail extension **186** projecting from the base **14**. The combination of the rails **182** and rail extensions **186** provide protection against luggage damage when the luggage bag **10** is pulled over a curb or other uneven surface by providing a bumper that continuously or directionally extends from the base **14** along a portion of the cover **18**, while also facilitating collapsibility of the luggage bag **10** through disengagement of the rails **182** from the rail extensions **186**. While the illustrated embodiment depicts the plurality of rails **182** and the plurality of corresponding rail extensions **186** as three rails **182** and three rail extensions **186**, in other embodiments, the plurality of rails **182** may include any number of rails **182** (e.g., two to six or more) and the plurality of rail extensions **186** may include any corresponding number of rail extensions **186** (e.g., two to six or more).

[0087] Referring to FIGS. **6**A-**6**B, the rails **182** are illustrated as a plurality of rails **182***a, b, c* positioned on a panel or bottom back panel **86** within the second panel zone **50***b*. Each rail **182***a, b, c* is coupled to the panel **86** by one or more attachment members, illustrated as a plurality of rivets through each rail **182***a, b, c*. In other embodiments, the rails **182***a, b, c* can be attached to the panel **86** by any suitable attachment member. Each rail **182***a, b, c* defines a channel **190** (best shown in FIG. **6**C). The channel **190** may further be defined by an elongated portion **194** of the rail **182***a, b, c* (shown in FIGS. **6**B and **6**D), which is integrally formed with the rail **182***a, b, c*. The rails **182***a, b, c* extend from the panel **86** in the second panel zone **50***b* towards the base **14**, crossing the fold **42** between the first and second panel zones **50***a, b* into the first panel zone **50***a*. More specifically, the elongated portion **194** of each rail **182***a, b, c* crosses the fold **42** between the first and second panel zones **50***a, b* and into the first panel zone **50***a*. Each rail **182** is approximately 1.9 inches wide, but in other embodiments may be anywhere from 1.5 inches to 3.0 inches or more wide.

[0088] Referring to FIGS. **6**A-**6**D, the rail extensions **186** are illustrated as a plurality of rail extensions **186***a, b, c* that are positioned on and project away from the base **14**. In the embodiment shown, the rail extensions **186***a, b, c* are each generally in alignment with and project towards the corresponding rails **182***a, b, c*. The outermost rail extensions **186***a, c* are further generally aligned with respective wheels **136***a, b* of the base **14**, while the intermediate rail extension **186***b* is between, and may be centered between, the outermost rail extensions **186***a, c*.

[0089] Each rail extension **186***a, b, c* includes a projection **198***a, b, c* (shown in FIGS. **6**C-**6**D). Each projection **198** is approximately 1.5 inches wide, but in other embodiments may be anywhere from 1.2 inches to 2.0 inches or more wide, and more specifically may be any width that is complementary to an associated rail **182** in order to facilitate engagement between the rail **182** and the projection **198**. Each rail extension **186** extends approximately 1.8 inches from the base **14**, with the projection **198** being approximately 0.6 inches (or 33% of the rail extension **186** length). In other embodiments, the length of the rail extension **186** may be any suitable length, and the corresponding length of the projection **198** may be any suitable length or percentage of the rail extension **186** length.

[0090] Each of the rails **182***a, b, c* removably engages a corresponding rail extension **186***a, b, c*. More specifically, each projection **198***a, b, c* is removably received by the elongated portion **194** of the channel **190** of a corresponding rail **182***a, b, c* to form the bumper **180**, which continuously extends from the base **14** along a portion of the cover **18** when the luggage bag **10** is in the upright position illustrated in FIGS. **6**A-**6**B (or a closed configuration, as shown in FIG. **1**). While the illustrated embodiment discloses the removable engagement in the form of the rails **182***a, b, c* receiving a portion of a corresponding rail extension **186***a, b, c*, in other embodiments any suitable removable engagement between the rails **182** and rail extensions **186** can be used. For example, the rail extensions **186** may alternatively define a respective channel having a size suitable to receive a portion of a corresponding rail **182**. As another example, each rail **182** may removably couple to a corresponding rail extension **186** by a connection member (e.g., a snap button, an interference fit, or other suitable fastener). In yet another embodiment, a combination of removable engagement as illustrated in FIGS. **6**A-**6**D together with the use of one or more connection members may be employed.

[0091] To facilitate collapsibility of the luggage bag **10**, each of the rails **182***a, b, c* disengages a corresponding rail extension **186***a, b, c* when the luggage bag **10** is in a collapsed, folded position (or a collapsed configuration) as illustrated in FIGS. **6**C-**6**D. To disengage each rail **182***a, b, c* from the corresponding rail extension **186***a, b, c*, the user collapses the panel **86** having the attached rails **182***a, b, c* along the fold **42**. The rails **182***a, b, c* disengage from the associated rail extensions **186***a, b, c*, as the rails **182***a, b, c* separate from the associated projections **198***a, b, c* during collapse of the cover **18**. In other embodiments, the rails **182***a, b, c* may disengage from the associated rail extensions **186***a, b, c* by sliding, pivoting, lifting away, or otherwise through any suitable manner of separating the rails **182***a, b, c* from the rail extensions **186***a, b, c*.

[0092] Referring back to FIG. **1**, in some embodiments, the luggage bag **10** includes a storage pocket **90** on the front portion **22**, more specifically on a panel or bottom front panel **94** within the second panel zone **50***b*, adjacent but separate from the base **14**. Referring to FIG. **27**A, in some embodiments, the storage pocket **90** is located on the front portion **322**, above the sub panel **347**, as discussed in more detail below. The storage pocket **90** includes a zipper, zip fastener, or any other suitable fastener to permit access to the inside of the storage pocket **90**. In other embodiments, the storage pocket **90** can be located on another panel on the front, back, first side, or second side portions **22**, **26**, **30**, **34**. In addition, more than one storage pocket **90** may be located on the luggage bag **10**. To facilitate collapsibility of the luggage bag **10**, the storage pocket **90** is arranged on a single panel, and does not extend to any adjacent panel or to the base **14**.

[0093] Referring now to FIG. **8**, in some embodiments, the back portion **26** is integrally formed with the first side portion **30** and second side portion **34**. For example, the back, first side, and second side portions **26**, **30**, **34** can be stitched together or otherwise connected or formed as a unitary portion of cover **18**. In some embodiments, a connection member **98** (see also FIGS. **4**A-B) in the form of a zip fastener or zipper separates the front portion **22** from the first and second side portions **30**, **34** and the back portion **26**. In such an embodiment, a first portion of the connection member **98** defines a portion of a perimeter of the front portion **22**, while a second portion of the connection member **98** defines a portion of a perimeter of the first and second side portions **30**, **34**, and extends along the arcuate edges **58***a*, **58***b* to the end edge **54** of the back portion **26**. The connection member **98** thus removably connects the front portion **22** to the back, first side, and second side portions **26**, **30**, **34**. In some embodiments, the connection member **98** further separates the first and second side portions **30**, **34** from the base **14**. In such an embodiment, the connection member **98** thus removably connects the front portion **22** to the back, first side, and second side portions **26**, **30**, **34**, and further removably connects the first and second side portions **330**, **334** from the base **314**.

[0094] It should be appreciated that the connection member **98** may be any suitable device or assembly for connecting the front portion **22** to the back, first side, and second side portions **26**, **30**, **34**, including, but not limited to, a hook and loop fastener, a zip fastener, or a fly fastener. In addition, while the connection member **98** is illustrated in FIGS. **44**A-B with two, opposing sliders **100***a*, **100***b* (such as in a two-way or a double-separating zip fastener), in other embodiments the connection member **98** may include one slider, or three or more sliders.

[0095] When the front portion **22** is connected to or engaged with the back, first side, and second side portions **26**, **30**, **34** (as shown in FIG. **1**), the portions **22**, **26**, **30**, **34** define an interior chamber **102** for receiving items for transport. The front portion **22** may be selectively or partially disconnected or disengaged from the back, first side, and second side portions **26**, **30**, **34** so that a user has various degrees of access to the interior chamber **102**. In FIG. **8**, the luggage bag **10** is shown in the upright position with only a part of the front portion **22** connected or engaged with the first side and second side portions **30**, **34** by one or more connection members **98** (hereinafter referred to as connection member **98**), thereby partially exposing the interior chamber **102**. By partially exposing the interior chamber **102**, a user may place and position items into the interior chamber **102** while the luggage bag **10** remains in the upright position (or is otherwise free standing).

[0096] As illustrated in FIG. **8**, the luggage bag **10** is provided in the partial access or first standing access position. The front portion **22** is illustrated as bent over the fold **42** provided between panels in the second and third panel zones **50***b*, **50***c*. It should be appreciated that the front portion **22** may be selectively or partially disconnected from the back, first side, and second side portions **26**, **30**, **34** as to allow any desired number of panel zones **50** to bend or fold over a desired fold **42** in the front portion **22** and thereby provide varying amounts of access to the interior chamber **102**.

[0097] Referring now to FIG. **9**, in some embodiments, components of the luggage bag **10** also fold or bend to form a butterfly opening, or second opening access position, in order to provide greater access to the interior chamber **102**. In some embodiments, with the luggage bag **10** placed with the back portion **26** positioned on a floor or other supporting surface, the connection member **98** may be opened so that the front portion **22** is no longer connected with the back, first side, and second side portions **26**, **30**, **34**, but remains connected to the base **14** to expose the interior chamber **102**. In other embodiments, the connection member **98** may be opened so that the front portion **22** is no longer connected with the back, first side, and second side portions **26**, **30**, **34**, and the first and second side portions are no longer connected with the base **14**.

[0098] In some embodiments, the first and second side portions **30**, **34** remain attached to the base **14**. In these embodiments, the first and second side portions **30**, **34** are then free to pivot or fold away from each other along the respective folds **42** (or seams) between the side portions **30**, **34** and the back portion **26**. As the first and second side portions **30**, **34** fold away from each other, the portions **30**, **34** are each at an oblique angle to the back portion **26**. The panels of the side portions **30**, **34** in the second panel zone **50***b* also fold about the angled folds **42***a*.

[0099] In some embodiments, the first and second side portions **330**, **334** are detached from the base **14**, as illustrated in FIG. **29**. In these embodiments, as the first and second side portions **330**, **334** fold away from each other, the portions **330**, **334** can lie flat in a plane with the back portion **326**. Once one or more items are placed into the interior chamber **102**, the side portions **30**, **34** are pivoted or folded towards each other, and the connection member **98** is reconnected (or closed), securing the front portion **22** to the back, first side, and second side portions **26**, **30**, **34**, and optionally securing the side portion **30**, **34** to the base **14**, thereby closing the interior chamber **102**.

[0100] This butterfly opening allows for unobstructed insertion of large or bulky items into the interior chamber **102**, as not only do the side portions **30**, **34** pivot or fold away from each other, with the arcuate face **66** removed there is no lip or other edge structure that would obstruct or otherwise hinder insertion of items into the interior chamber **102**. In other words, when the travel luggage bag is closed or opened in the manner shown in FIG. **8**, the arcuate face **66** overlaps a portion of the base **14** to enclose the interior chamber **102**. By opening the arcuate face **66** away from the side portions **30**, **34** and back portion **26**, and then pivoting or folding the side portions **30**, **34** away from each other along folds **42**, the back, first side, and second side portions **26**, **30**, **34** thereby define a butterfly opening that provides ready access to the interior chamber **102** unobstructed by a portion of the cover **18**. In other embodiments, the front portion **22** may be detachable or otherwise removable from the base **14**.

[0101] With reference to FIGS. **10-15**, the base **14** is illustrated in greater detail. The base **14** includes a bottom external surface **110** opposite an inside surface **114**. The inside surface **114** is defined or surrounded by a front wall **122** opposite a back wall **126**, and a first side wall **130** opposite a second side wall **134**. The front wall **122** is located on the same side of the luggage bag **10** as the front portion **22**, while the back wall **126** is provided on the same side of the luggage bag **10** as the back portion **26**. As illustrated in FIGS. **14** and **15**, the first and second side walls **130**, **134** increase in height from the front wall **122** to the back wall **126**, with the height being the distance of the wall away from the bottom surface **110**. In other embodiments, the first and second side walls **130**, **134** may have a uniform height from the front wall **122** to the back wall **126**. The inside surface **114** preferably defines a planar or substantially flat receiving surface substantially free of protrusions or other structural obstructions that may interfere with receipt of items into the inside surface **114** of the base **14**.

[0102] The base **14** includes a pair of wheels or skate wheels **136***a*, **136***b* provided on the back wall **126** side. The skate wheels **136***a*, **136***b* do not swivel about the base **14**, and extend beyond a plane defined by the back wall **126** such that a portion of each of the wheels **136***a*, **136***b* extends outside of the base footprint defined by the bottom surface **110** and walls **122**, **126**, **130**, **134**. The wheels **136***a*, **136***b* include a common axis of rotation **138** preferably offset from the bottom surface **110** and a plane defined by the back wall **126** and are configured to act as a fulcrum about which the luggage bag **10** pivots from the upright position (see FIGS. **1** and **8**) to a tilted position (see FIG. **19**). The bottom surface **110** also includes a curved portion **139** between the wheels **136***a*, **136***b* (shown in FIGS. **10** and **13**) defined by a radius preferably in a range of 50 mm to 70 mm, and more preferably about 60 mm. The radius of the curved portion **139** provides a ground clearance zone between the wheels **136***a*, **136***b* and the bottom surface **110**.

[0103] The base **14** further includes a pair of caster wheels **142***a*, **142***b*, which swivel about the base **14**. The caster wheels **142***a*, **142***b* each preferably swivel 360 degrees about the base **14** around a swivel axis to allow the luggage bag **10** to roll in a controlled manner when in the upright position.

[0104] As illustrated in FIG. **10**, the wheels **136***a*, **136***b* are preferably separated by a first distance D_1 of approximately 330 mm between a point of rotation of each of the wheels **136***a*, **136***b* while the caster wheels **142***a*, **142***b* are preferably separated by a second distance D_2 of approximately 300 mm between a swivel axis of each of the wheels **142***a*, **142***b*. In other embodiments, the first and second distances D_1, D_2 can be any suitable or desired distance, with the first distance D_1 generally being greater than the second distance D_2. The wheels **136**, **142** are positioned in the same plane, i.e., the wheels **136**, **142** are positioned to contact a floor or other surface in the same horizontal plane. When the luggage bag **10** is in the upright position, both the non-swivel wheels **136***a*, **136***b* and the swivel wheels **142***a*, **142***b* remain in contact with the floor or other surface on which the luggage bag **10** is located. This contact by all wheels **136**, **142** on the base **14** reduces the risk of unintended luggage movement while continuing to allow targeted rolling movement of the luggage bag **10**. While the swivel wheels **142** permit directional movement of the luggage bag **10**, the non-swivel wheels **136** act as a stop to help inhibit unintended luggage movement, for example if a person or object pushes (or applies a force) on one of the side portions **30**, **34**.

[0105] The luggage bag **10** is collapsible to reduce the storage footprint when not in use. The panels of the cover **18** fold along folds **42**, **42***a* allowing the panels to be collapsed toward and at least partially received in the base **14**. As referenced earlier, folds **42** are provided between adjacent or consecutive panels in each of the front, back, first side, and second side portions **22**, **26**, **30**, **34**. In addition, folds **42** are provided between panels of each adjacent or consecutive portion **26**, **30**, **34**, such as between adjacent panels in a given panel zone **50**. In some embodiments, the angled folds **42***a*, which are generally oblique to the base **14**, also serve to facilitate collapsing luggage bag **10** toward base **14**.

[0106] In some embodiments where the side portions **30**, **34** remain attached to the base **14**, to collapse the luggage bag **10**, the connection member **98** is opened such that the front portion **22** is no longer secured to the back, first side, and second side portions **26**, **30**, **34**. Referring to FIGS. **16** and **17**, in these embodiments, the front portion **22** is folded or bent along the folds **42** between panels, and then received in the base **14**. The side portions **30**, **34** are also folded or bent along angled folds **42***a* towards the base **14**. The panel between the base **14** and folds **42***a* is then received by the base **14**, followed by the panel on the opposite side of folds **42***a*. With the side portions **30**, **34** positioned inward (or toward each other) to contact the back portion **26**, if used, the rails **182** disengage from the rail extensions **186**, and the remainder of the cover **18** folds along folds **42** and is partially received in the base **14** (see FIG. **17**). In other embodiments, the cover **18** may be partially, substantially, or entirely received in the base **14**.

[0107] In other embodiments, such as the luggage bag **300**, the side portions **330**, **334** are detached from the base **14**, as discussed in more detail below. In such an embodiment, to collapse the luggage bag **300**, the connection member **98** is opened such that the front portion **322** is no longer secured to the back, first side, and second side portions **326**, **330**, **334**, and further opened such that the side portions **330**, **334** are no longer secured to the base. Referring to FIGS. **31**A-**31**C, in these embodiments, the front portion **322** is first folded along a sub panel **342***b* and further folded along the folds **342** between panels **346**, and then received in the base **314**. The side portions **330**, **334** are folded inwards towards the back portion **326** and then fully received by the base **314**. To further secure the cover **318** into the base **314**, the luggage bag **300** includes a clip **371** attached to the base **314** that is received within a loop attached to the back portion **326**, as illustrated in FIG. **30**.

[0108] In some embodiments, in this collapsed, folded position, the luggage bag **10** itself may be received in a storage bag **106** (see FIGS. **18**A-B) for storage until later use. While the cover **18** remains in the collapsed, folded position by way of the illustrated arrangement of panels and folds, the luggage bag **10** may include additional devices or assemblies to assist in retaining the luggage bag **10** in the collapsed, folded position for orderly storage. In some embodiments, the luggage bag **10** can include a simple strap or other device to maintain the cover **18** in the collapsed position and avoid an unintentional unraveling of the cover **18** from the collapsed position (for example by an unintended dropping of the collapsed, folded luggage bag **10**). Such a simple strap may include a single strap or bungee-like cord that extends around a portion of the base **14** and cover **18** to assist in maintaining the cover in the collapsed, folded position.

[0109] FIG. **26** illustrates another embodiment of a storage bag **160** for receiving the luggage bag **10** when the luggage bag **10** is in a collapsed or folded position. The storage bag **160** comprises a front panel **161**, a back panel **162**, a first side panel **163**, a second side panel **164**, a bottom panel **165**, and two handles **166**. The front panel **161**, the back panel **162**, the first side panel **163**, the second side panel **164**, and the bottom panel **165** each comprise a rectangular or approximately rectangular shape. Together, the panels form a box-like shape. The front, back, first side, and second side panels **161**, **162**, **163**, and **164** are aligned perpendicular to the bottom panel **165**. The first and second side panels **163** and **164** are aligned perpendicular to the front and back panels **161** and **162**. A top of the storage bag **160** comprises a perimeter formed by edges of the front, back, first side, and second side panels **161**, **162**, **163**, and **164**.

[0110] The storage bag **160** comprises a height, measured orthogonal to the bottom panel from the bottom panel to the top perimeter of the storage bag **160**. The height can range from 30 to 60 cm. In some embodiments, the height ranges from 30 to 40 cm, 35 to 45 cm, 40 to 50 cm, 45 to 55 cm, or 50 cm to 60 cm. In some embodiments, the height can be 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, 48 cm, 49 cm, 50 cm, 51 cm, 52 cm, 53 cm, 54 cm, 55 cm, 56 cm, 57 cm, 58 cm, 59 cm, or 60 cm. The storage bag **160** comprises a width, measured parallel to an intersection between the bottom panel **165** and the front panel **161**. The width can range from 35 to 55 cm. In some embodiments, the width ranges from 35 to 45 cm, 40 to 50 cm, or 45 to 55 cm. In some embodiments, the width can be 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, 48 cm, 49 cm, 50 cm, 51 cm, 52 cm, 53 cm, 54 cm, or 55 cm. The storage bag **160** comprises a depth, measured parallel to an intersection between the bottom panel **165** and the first side panel **163**. The depth ranges from 30 to 50 cm. In some embodiments, the depth ranges from 30 to 40 cm, 35 to 45 cm, or 40 to 50 cm. In some embodiments, the height can be 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, 48 cm, 49 cm, or 50 cm.

[0111] The storage bag **160** can further comprise handles **166**. In some embodiments, a first handle is connected to the first side panel **163**, and a second handle is connected to the second side panel **164**. Additionally, in some embodiments, the top perimeter of the storage bag comprises a channel for receiving and retaining a drawstring. In these embodiments, the drawstring extends through an opening in the channel adjacent the front panel **161**.

[0112] Each panel **161**, **162**, **163**, **164** and **165** of the storage bag **160** can be formed of a polyethylene board material $[(C_2H_2)_nH_2]$, a foam material, a cloth material, or a combination thereof. The polyethylene board material may be any industry standard grade, including, but not limited to, ultra-high molecular weight polyethylene (UHMWPE), ultra-low molecular weight polyethylene (ULMWPE), high molecular weight polyethylene (HMWPE), high-density polyethylene (HDPE), high-density cross-linked polyethylene (HDXLPE), cross-linked polyethylene (PEX), medium-density polyethylene (MDPE), linear low-density polyethylene (LLDPE), low-density polyethylene (LDPE), very-low density polyethylene (VLDPE), and chlorinated polyethylene (CPE). In other embodiments, the panels **161**, **162**, **163**, **164** and **165** may be formed of polyurethane, acrylonitrile butadiene styrene, combinations thereof, or any other suitable material. The composition of the storage bag **160** provides enough flexibility for the drawstring mechanism to partially collapse the storage bag **160** and hold in the collapsed luggage bag **10**. However, the panels **161**, **162**, **163**, **164** and **165** of the storage bag **106** also provide a level of rigidity that allows the luggage bag to be stored in a box-shaped container.

[0113] Referring now to FIG. **19**, in many embodiments the luggage bag **10** reduces strain on a user when in a tilted position or pivot position for rolling movement on the wheels **136**. To reach the illustrated tilted position, a user pivots the luggage bag **10** about the axis of rotation **138** of wheels **136**, for example with rolling handle **74**. During the pivot, the front wall **122** end of the base **14** lifts away from the ground or surface **146**. In addition, all swivel wheels **142** are lifted away from contact with the surface **146**. The user then pulls on the handle **74**, and the luggage bag **10** rolls on wheels **136**. The positioning of the handle **74** on the back portion **26** advantageously reduces strain on a wrist, hand, and arm of the user pulling (or applying a pulling force to) the luggage bag **10** as the handle **74** location is further away or offset from the wheels **136** in a pulling direction, reducing the force A_f applied by the user to overcome the load force L_f of the luggage bag **10**. In addition, a portion of the luggage bag **10** rests on the hand, wrist, and/or arm of the user as the user pulls the luggage bag **10** in the tilted position, reducing the application of luggage weight to a user's hand, wrist, and arm.

[0114] Referring now to FIGS. **24** and **25**, in some embodiments of the luggage bag **10**, the interior chamber **102** further comprises an internal cover **150**. The internal cover **150** can serve to further add protection and padding around golf club heads that are stored within the luggage bag **10**. The internal cover **150** is shaped to fit within a top portion of the interior chamber **102** of the luggage bag **10**. The internal cover **150** comprises a front panel **151**, a first side panel **153**, a second side panel **154**, and a back panel **152**. The front panel **151** curves to also form a top of the internal cover **150**. The internal cover **150** further comprises an opening generally facing towards the base of the luggage bag **10**. The internal cover **150** is attached to the luggage bag **10** only along one seam of the internal cover. The attached seam **155** is between the front panel and the rear panel, adjacent the top of the internal cover **150**. The back panel **152** of the internal cover **150** can comprise a height, measured vertically, between 15 cm and 30 cm. For example, the back panel height of the internal cover can be 15 cm, 16 cm, 17 cm, 18 cm, 19 cm, 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, or 30 cm. In the illustrated embodiment, the height of the back panel **152** is approximately 22 cm. The first and second side panels **153**, **154** can each comprise a depth, measured parallel to the ground plane and roughly orthogonal to the back panel **152**. The first side panel depth equals the second side panel depth. The depth of the side panels **153**, **154** can range between 10 cm and 25 cm. For example, the depth of the side panels **153**, **154** can be 10 cm, 11 cm, 12 cm, 13 cm, 14 cm, 15 cm, 16 cm, 17 cm, 18 cm, 19 cm, 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, or 25 cm. In the illustrated embodiment, the depth of the side panels **153**, **154** is approximately 16 cm. The front and back panels **151**, **152** can each comprise a width, measured parallel to the ground plane and roughly orthogonal to the side panels **153**, **154**. In most embodiments, the front and back panel widths are the same. The front and back panel widths can range between 15 cm and 30 cm. For example, the front and back panel widths can be 15 cm, 16 cm, 17 cm, 18 cm, 19 cm, 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, or 30 cm. In the illustrated embodiment, the front and back panel width is approximately 22 cm.

[0115] To prevent rattling and shifting of the golf club heads, the opening of the internal cover **150** comprises a perimeter draw string mechanism **156**. The draw string mechanism **156** can comprise a string, a cord, a rope, a cord lock, a hem, a casing, loops, or other draw string mechanism elements known in the art. The perimeter draw string mechanism **156** secures the internal cover **150** around the golf club heads. In some embodiments, the internal cover

150 comprises fabric, foam, mesh, or other suitable padding material. In many embodiments, the internal cover **150** can comprise a thickness ranging between 0.05 cm and 2.0 cm, or between 0.5 and 1.5 cm. For example, the internal cover **150** can comprise a 1 cm thick open cell foam. In some embodiments, stitching is sewn across portions of the panels **151**, **152**, **153**, and **154**.

I. First Exemplary Embodiment

[0116] FIGS. **20**-**23** illustrate a first exemplary embodiment of the rolling collapsible luggage bag described herein. The example luggage bag **200** is similar to the luggage bag **10**. In this embodiment, a first panel zone **250***a*, a second panel zone **250***b*, and a third panel zone **250***c* each comprise shapes relatively similar to the first panel zone **50***a*, second panel zone **50***b*, and third panel zone **50***c*, respectively, of the luggage bag **10**. However, the fourth **250***d* and fifth panel **250***e* zones of the luggage bag **200** comprise a slightly different shape than the fourth panel zone **50***d* and fifth panel zone **50***e* of the luggage bag **10**. From a front or rear view, illustrated in FIGS. **20** and **21**, the fourth **250***d* and fifth panel **250***e* zones both comprise trapezoidal shapes that taper more sharply than in the first described embodiment. In other words, a fourth width W_4 of luggage bag **200** is less than the fourth width W_4 of luggage bag **10**.

[0117] In the illustrated embodiment, the height of the first panel zone **250***a* is approximately 6 cm, the height of the second panel zone **250***b* is 31.5 cm, the height of the third panel zone **250***c* is approximately 31.5, the height of the fourth panel zone **250***d* is approximately 31.5 cm, and the height of the fifth panel zone **250***e* is approximately 21.5 cm, when the heights are measured along a back portion **226**. Additionally, from a side view, as illustrated in FIGS. **22** and **23**, the fourth panel zone **250***d* can comprise a side edge with an angle change roughly halfway up the height of the panel. A top of the side edge of the leads into the fifth panel zone, which has a roughly straight angled edge instead of an arcuate edge. The edge of the fifth panel zone also comprises a change in angle in order to provide a small flat top portion that engages a handle.

II. Second Exemplary Embodiment

[0118] FIGS. **27**A-**33** illustrate another exemplary embodiment of the rolling collapsible luggage bag described herein. The luggage bag **300** includes a cover **318** that lies flat in the opened configuration to provide full access to the interior chamber **304**. The luggage bag **300** can provide maximum access to the interior chamber **304** without compromising the ability to fully collapse. The luggage bag **300** is similar to the luggage bag **10** such that the luggage bag **300** includes a front flap **322**, which is fully detachable from the back, first side, and second side portion **326**, **330**, **334**. However, the first and second side portions **330**, **334** are configured to fully detach from the base **314** to form a flat surface for a user to insert items into the luggage bag **300**. Therefore, the luggage bag **300** provides the maximum amount of access to the interior chamber **304**. The detachability of the side portions **330**, **334** provides further benefits to the luggage bag **300** such as improved collapsibility and storage, as discussed in more detail below.

[0119] Referring to FIG. **27**A, the front flap **322** is integrally formed with the base **314** and selectively removable from the back, first side, and second side portions **326**, **330**, **334** to provide access to the interior chamber **304**. Further, the back portion **326** is integrally formed with the base, first side, and second side portions **314**, **330**, **334**, similar to the luggage bag **10**. The luggage bag **300** differs from the luggage bag **10** in that the first side portion **330** and the second side portion **334** are selectively removable from the base **314**. When lying flat in the open configuration, the back, first side, and second side portions **326**, **330**, **334** form a flat surface that provides full access to the interior chamber **304**, as illustrated in FIG. **29**. In many previous designs, the back and side portions were not fully removable from the base, and therefore, did not provide maximum access to the interior chamber.

[0120] The luggage bag **300** comprises a plurality of panels **346** arranged into a plurality of panel zones **350**. The construction and composition of the panels **346** of the luggage bag **300** are similar to the panels **46** of the luggage bag **10**. The panels **346** of the luggage bag **300** each define a front height near the front portion **322**, and a side height near the first and second side portions **330**, **334**. The height of the panels **346** within each zone **350** can be similar, or the height can vary. The main difference in the exemplary luggage bag **300** is the arrangement of the panels **346** within the panel zones **350**. For example, in some panel zones **350**, the panels **346** are stacked such that the panels **346** within the front portion, the back portion, and the first side, and second side portions **322**, **326**, **330**, **334** are offset from one another, as illustrated in FIG. **27**A. The offset of the panels allows the cover **318** to fully collapse into the base **314**.

[0121] Referring to FIGS. **28**A and **28**B, the panels **346***a* within the first panel zone **350***a* have a first front panel height H_{1F} of approximately 11 cm, but as discussed above, the first panel height H_1 can vary depending on the size, shape, and/or orientation of the luggage bag **300**. The panels **346***a* within the first panel zone **350***a* further have a first side panel height His that decreases from the front to the rear such that the panels **346***a* are taller near the front flap **322** and shorter near the back portion **326** to conform to the sloping height of the base **314**, as discussed in more detail below. The first side panel height H_{S1} can be approximately 5.0 cm, 5.5 cm, 6.0 cm, 6.5 cm, 7.0 cm, 7.5 cm, 8.0 cm, 8.5 cm, 9.0 cm, 9.5 cm, 10.0 cm, 10.5 cm, 11.0 cm, 11.5 cm, 12 cm, 12.5 cm, 13.0 cm, 13.5 cm, 14.0 cm, 14.5 cm, or 15.0 cm. In some embodiments, the first side panel height H_{S1} can be less than 6.5 cm, 6.4 cm, 6.3 cm, 6.2 cm, 6.1 cm, or 6.0 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the first side panel height H_{S1} can be greater than 6.5 cm, 6.6 cm, 6.7 cm, 6.8 cm, 6.9 cm, or 7.0 cm based on the size, shape, or orientation of the luggage bag **10**. In some embodiments, the first side panel height His tapers from between approximately 11 cm near the front flap **322** to approximately 2 cm near the back portion **326**. The taper in the panels **346***a* from near the front flap **322** to near the back portion **326** creates triangular shaped panels near the first and second side portions **330**, **334**.

[0122] As discussed above, the luggage bag **300** utilizes a different collapsibility mechanism than the luggage bag **10**. The main difference is in the configuration of the panels **346***b* within the second panel zone **350***b*. Referring to FIG. **28**A, the panel **346***b* within the second panel zone **350***b* of the front flap **322** (the "main front panel **346***b*) comprises a sub panel **347**. The sub panel **347** is located closer to the first panel zone **350***a* than the main front panel **346***b*. A sub fold

342*b* defines a boundary between the sub panel **347** and the main front panel **346***b*. To collapse the luggage bag **300**, the front portion is first folded along the sub fold **342***b* such that the sub panel **347** is first received within the base **314**.

[0123] Referring to FIG. **28**A, the main front panel **346***b* defines a second main panel height H_{2MF} of approximately 25 cm. In some embodiments, the second main panel height H_{2MF} can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, or 40 cm. In some embodiments, the second main panel height H_{2MF} can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the second main panel height H_{2MF} can be greater than 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. The sub panel **347** defines a sub panel height H_{2SF} of approximately 12 cm. In some embodiments, the sub panel height H_{2SF} can be approximately 5.0 cm, 5.5 cm, 6.0 cm, 6.5 cm, 7.0 cm, 7.5 cm, 8.0 cm, 8.5 cm, 9.0 cm, 9.5 cm, 10.0 cm, 10.5 cm, 11.0 cm, 11.5 cm, 12 cm, 12.5 cm, 13.0 cm, 13.5 cm, 14.0 cm, 14.5 cm, or 15.0 cm. In some embodiments, the sub panel height H_{2SF} can be less than 6.5 cm, 6.4 cm, 6.3 cm, 6.2 cm, 6.1 cm, or 6.0 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the sub panel height H_{2SF} can be greater than 6.5 cm, 6.6 cm, 6.7 cm, 6.8 cm, 6.9 cm, or 7.0 cm based on the size, shape, or orientation of the luggage bag **10**.

[0124] Referring again to FIG. **28**A, the main front panel **346***b* within the second panel zone **350***b* and the sub panel **347** define a combined second front panel height H_{2F} of approximately 40 cm. In some embodiments, the second front panel height H_{2F} can be less than 50 cm. In some embodiments, the second front panel height H_{2F} can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, 48 cm, 49 cm, or 50 cm. In some embodiments, the second front panel height H_{2F} can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the second front panel height H_{2F} can be greater than 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**.

[0125] The sub panel height H_{2SF} is approximately the same as the height of the base **314**, thereby allowing the sub panel **347** to fold into the base **314** and rest against a front wall of the base, as illustrated in FIG. **31**A. The sub panel **347** raises the height of the main front panel **346***b*, such that the main front panel **346***b* is located at an offset from the panels **346***b* near the first and second side portions **330**, **334**, as illustrated in FIG. **27**A.

[0126] In addition to the sub panel **347**, the panels **346***b* within the first and second side portions **330**, **334** further differ from those of the luggage bag **10**. Specifically, the panels **346***b* do not include an angled fold as in the previous embodiments (see FIG. **6**). Instead, the panels **346***b* near the first and second side portions **330**, **334** each comprise a single panel. The single panel can prevent buckling within the second panel zone **350***b*, thereby preventing wear within the panels **346***b* at the intersection between the base **314** and the cover **318**. Wearing occurred in a similar area of the panels in a previous design, where the side panels were not fully removable from the base. In other embodiments, the angled fold can help facilitate the collapsibility of the cover. The luggage bag **300**, however, utilizes a different collapsibility mechanism, as discussed in more detail below.

[0127] Referring to FIG. **28**B, the panels **346***b* within the second panel zone **350***b* further have a second side panel height H_{2S} that increases from the front to the rear such that the panels **346***b* are taller near the back portion **326** and shorter near the front flap **322**. In some embodiments, the second side panel height H_{2S} can be less than 50 cm. In some embodiments, the second side panel height H_{2S} can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, 40 cm, 41 cm, 42 cm, 43 cm, 44 cm, 45 cm, 46 cm, 47 cm, 48 cm, 49 cm, or 50 cm. In some embodiments, the second side panel height H_{2S} can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the second side panel height H_{2S} can be greater than 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**. In one exemplary embodiment, the second side panel height H_{2S} increases from approximately 30 cm near the front flap **322** to approximately 36.5 cm near the back portion **326**.

[0128] The taper of the panels **346***b* within the second panel zone **350***b* follows the contour of the first panel zone panel **350***a* near the first and second side portions **330**, **334**. Referring to FIG. **27**A, the height increase in the panels **346***b* of the second panel zone **350***b* creates trapezoidal panels near the first and second side portions **330**, **334**.

[0129] As discussed above, the luggage bag **300** comprises a first panel zone **350***a* and a second panel zone **350***b* that differ from the first panel zone **50***a* and a second panel zone **50***b* of the luggage bag **10**. The luggage bag **300** further comprises a third panel zone **350***c*, a fourth panel zone **350***d*, and a fifth panel zone **350***e* each comprise shapes relatively similar to the first panel zone **50***a*, second panel zone **50***b*, and third panel zone **50***c*, respectively, of the luggage bag **10**.

[0130] Referring to FIGS. **28**A and **28**B, the panels **346***c* within the third panel zone **350***c* have a third front panel height H_{3F} of approximately 25 cm. In some embodiments, the third front panel height H_{3F} can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, or 40 cm. In some embodiments, the third front panel height H_{3F} can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the third front panel height H_{3F} can be greater than 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. The panels **346***c* within the third panel zone **350***c* further have a third side panel height H_{3S} of approximately 33.5 cm. In some embodiments, the third side panel height H_{3S} can be 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, 35 cm, 36 cm, 37 cm, 38 cm, 39 cm, or 40 cm. In some embodiments, the third side panel height H_{3S} can be less than 40 cm, 39 cm, 38 cm, 37 cm, 36 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the third side panel height H_{3S} can be greater than 30 cm, 31 cm, 32 cm, 33 cm, 34 cm, or 35 cm based on the size, shape, or orientation of the luggage bag **10**. Similar to the panels **346***b* of the second panel zone **350***b*, the panel **346***c* near the front flap **322** is located at an offset from the panels **346***c* near the first and second side portions **330**, **334**, as illustrated in FIG. **27**A. The heights H_{3F}, H_{3S} remain substantially constant such that the panels **346***c* are rectangular.

[0131] Referring to FIGS. **28**A and **28**B, the panels **346***d* within the fourth panel zone **350***d* have a fourth front panel height H_{4F} of approximately 25 cm. The panels **346***d* within the fourth panel zone **350***d* further have a fourth side panel height H_{4S} of approximately 33.5 cm. Similar to the panels **346***b* of the second panel zone **350***b*, the panel **346***d* near the front flap **322** is located at an offset from the panels **346***d* near the first and second side portions **330**, **334**, as illustrated in FIG. **27**A. The heights H_{4F}, H_{4S} remain substantially constant such that the panels **346***d* are rectangular.

[0132] Referring to FIGS. **28**A and **28**B, the panels **346***e* within the fifth panel zone **350***e* have a fifth front panel height H_{5F} and a fifth side panel height H_{5S} of approximately 20 cm. In some embodiments, the fifth front panel height H_{5F} can be 10 cm, 11 cm, 12 cm, 13 cm, 14 cm, 15 cm, 16 cm, 17 cm, 18 cm, 19 cm, 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, or 30 cm. In some embodiments, the fifth front panel height H_{5F} can be less than 30 cm, 29 cm, 28 cm, 27 cm, 26 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. In other embodiments, the fifth front panel height H_{5F} can be greater than 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, or 25 cm based on the size, shape, or orientation of the luggage bag **10**. The panels **346***e* within the fifth panel zone **350***e* are substantially similar to the panels **46***e* within the fifth panel zone **50***e* of the luggage bag **10** with regard to panel type, positioning, and shape.

[0133] The luggage bag **300** is similar to the luggage bag **10**, but for the differences in the panel configuration, and panel heights. The panels **346** of the luggage bag **300** define similar widths, depths, and thicknesses as the panels **46** of the luggage bag **10**. The luggage bag **300** further includes a plurality of handles **370***a*, **370***b*, **370***c*, **370***d*, **370***e*, and **374** similar to the handles of the luggage bag **10**.

[0134] Referring to FIG. **27**A, the luggage bag **300** further comprises a storage pocket **390** located on the front portion **322**. The storage pocket **390** is located above the sub panel **347** on the main front panel **346***b*. To facilitate collapsibility of the luggage bag **300**, the storage pocket **390** remains within the main front panel **346***b* and does not extend to any adjacent panel. In other words, the storage pocket **390** does not cross over the folds **342**, and therefore, does not hinder the collapsibility of the luggage bag **300**.

[0135] In some embodiments, the luggage bag **300** further comprises one or more inner pockets formed near the interior chamber **304**. Referring to FIGS. **32** and **33**, the luggage bag **300** can comprise a first inner pocket **379** near the first side portion **330**, and a second inner pocket **381** near the second side portion **334**. The one or more inner pockets **379**, **381** can be formed from mesh or can be zipper pockets. In some embodiments, the one or more inner pockets **379**, **381** can extend over the folds **342** such that the inner pockets **379**, **381** extend into multiple panel zones **350**, as illustrated in FIGS. **32** and **33**. In other embodiments, the inner pockets **379**, **381** do not extend into multiple panel zones **350**, as illustrated in FIG. **29**. The inner pockets **379**, **381** can be accessed while the luggage bag **300** is in the open configuration.

[0136] FIG. **29** illustrates the luggage bag **300** in the opened configuration in which clubs can be stored. Similar to the luggage bag **10**, the back portion **326** is integrally formed with the first side portion **330**, the second side portion **334**, and the base **314**. However, in contrast to the luggage bag **10**, the first and second side portions **330**, **334** are not formed integrally with the base **314**. As such, the first side portion **330** defines a first bottom edge **355**, and the second side portion defines a second bottom edge **357**. The first and second bottom edges **355**, **357** each define a portion of the perimeter of the first and second side portions **330**, **334**, respectively. To open the luggage bag **300**, the connection member **398** (see also FIGS. **4**A-B) separates the front portion **322** from the first and second side portions **330**, **334** and the back portion **326**. The connection member **398** further separates the first and second side portions **330**, **334** from the base **314**. A first portion of the connection member **398** defines a portion of a perimeter of the front portion **322**, while a second portion of the connection member **398** defines a portion of a perimeter of the first and second side portions **330**, **334**, extends along the first and second bottom edges **355**, **357**. The connection member **398** thus removably connects the front portion **322** to the back, first side, and second side portions **326**, **330**, **334**, and further removably connects the first and second side portions **330**, **334** from the base **314**. Since the first and second bottom edges **355**, **357** are not formed integrally with the base **314**, they allow the first and second side portions **330**, **334** to open away from the base **314**. Similar to the luggage bag **10**, the front portion **322** of the luggage bag **300** can be selectively or partially disconnected or disengaged from the back, first side, and second side portions **326**, **330**, **334** so that a user has various degrees of access to the interior chamber **304**.

[0137] While the connection mechanism of the front flap **322** is similar to that of the luggage bag **10**, the luggage bag **300** includes an improved butterfly opening of the luggage bag **300**. Referring again to FIG. **29**, the first and second side portions **330**, **334** fold or bend to form a flat surface, or second opening access position, in order to provide greater access to the interior chamber **304**. With the luggage bag **300** placed with the back portion **326** positioned on a floor or other supporting surface, the connection member **398** can be opened so that the front portion **322** is no longer connected with the back portion **326**, the first side portion **330**, and the second side portion **334** to expose the interior chamber **304**. The connection member can further be opened so that the first and second side portions are no longer connected with the base **314**. The first and second side portions **330**, **334** are then free to pivot or fold away from each other along the respective folds **342** (or seams) between the side portions **330**, **334** and the back portion **326**.

[0138] In the open configuration, the first and second side portions **330**, **334** lie flat in a plane with the back portion **326**. This flat, mat-like opening allows for unobstructed insertion of large or bulky items into the interior chamber **304** as there is no lip or other edge structure that would obstruct or otherwise hinder insertion of items into the interior chamber **304**. Therefore, the luggage bag **300** provides the maximum allowable access to the interior chamber **304**.

[0139] Once one or more items are placed into the interior chamber **304**, the side portions **330**, **334** are pivoted or folded towards each other, and the connection member **398** is reconnected (or closed), securing the first and second side portions **330**, **334** to the base, and further securing the front portion **322** to the back, first side, and second side portions **326**, **330**, **334**, thereby closing the interior chamber **304**. In many prior art luggage bags, the cover is arranged with a single connection member or a zipper through a middle section. In these embodiments, the user has to maneuver a golf bag through a single, central opening of the luggage bag, and further maneuver the zipper around the golf bag to enclose the interior cavity. The luggage bag **300**, however, allows the user to place items on to a flat surface (the flattened back and first and second side portions **326**, **330**, **334**) and to easily reconnect each portion to enclose the interior chamber **304**.

[0140] The base **314** of the luggage bag **300** is similar to the base **14** of the luggage bag **10**. Specifically, the height of the base **314** increases from near the front portion **322** to near the back portion **326**. However, the base **314** can be slightly taller near the front portion **322** to facilitate collapsibility, as discussed in more detail below. In some embodiments, the height of the base can be 10 cm, 11 cm, 12 cm, 13 cm, 14 cm, 15 cm, 16 cm, 17 cm, 18 cm, 19 cm, 20 cm, 21 cm, 22 cm, 23 cm, 24 cm, 25 cm, 26 cm, 27 cm, 28 cm, 29 cm, or 30 cm. In one exemplary embodiment, the height of the base **314** is 16.2 cm near the front portion **322**.

[0141] The luggage bag **300** is collapsible to reduce the storage footprint when not in use. The panels **346** of the cover **318** fold along folds **342**, **342*b*** allowing the panels **346** to be collapsed toward and received in the base **314**. The folds **342** are provided between adjacent or consecutive panels in each of the front, back, first side, and second side portions **322**, **326**, **330**, **334**. In addition, folds **342** are provided between panels of each adjacent or consecutive portion **326**, **330**, **334**, such as between adjacent panels in a given panel zone **350**.

[0142] FIGS. **31**A-**31**C illustrate the collapsibility of the luggage bag **300**. The collapsibility of the luggage bag **300** slightly differs from that of the luggage bag **10**. First, the luggage bag **300** does not include an angled fold within the second panel zone **350*b*** to facilitate the collapsibility of the first and second side portions **330**, **334**. Rather, the front portion **322** of the luggage bag **300** further comprises a sub fold **342*b*** that allows the sub panel **347** to collapse inward toward the base **314**.

[0143] To collapse the luggage bag **300**, the connection member **398** is opened such that the front portion **322** is no longer secured to the back, first side, and second side portions **326**, **330**, **334**, and further opened such that the side portions **330**, **334** are no longer secured to the base. Referring to FIG. **31**A, the front portion **322** is first folded along the sub fold **342*b*** such that the sub panel **347** is the first panel received within the base **314**. The front portion **322** is further folded along the folds **342** between panels **346**, and then received in the base **314**. The side portions **330**, **334** are folded inwards towards the back portion **326** and received by the base **314**. If used, the rails **382** disengage from the rail extensions **386**, and the remainder of the cover **318** folds along folds **342** and is partially received in the base **314** (see FIGS. **31**A-**31**C).

[0144] The luggage bag **300** utilizes a self-storage function and does not require an additional storage bag to assist in retaining the luggage bag **300** in the collapsed, folded position for orderly storage. Referring to FIG. **30**, this self-storage function is facilitated by a clip **371** positioned on the base **314** that is received within a loop **375** positioned on the back portion **326**. In the collapsed position, the clip **371** engages the loop **375** to compress the cover **318** into the base **314**. The clip **371** is secured to the base via a first strap **373**, which is attached to the base **314** near the front of the luggage bag **300**. The loop **375** is secured to the back portion **326** via a second strap **377**, which is attached to the panel **346*b*** near the back portion, as illustrated in FIG. **27**B. The first and second straps **373**, **377** can be sewn, or otherwise attached to the luggage bag **300**. As discussed above, in the collapsed position, the cover **318** is received within the base such that the panel **346*b*** near the back portion **326** is exposed to the exterior. As such, in the collapsed position, the loop **375** is proximate the clip **372**, thereby allowing the user to easily hook the clip **371** within the loop **375** to compress the cover **318** within the base **314**. When the luggage bag **300** is in the collapsed position, the luggage bag **300** is movable along the ground or a supporting surface via the wheels. Therefore, the user can utilize the handle **370*a*** easily stow the luggage bag **300** in a low position.

Method of Manufacture

[0145] A method of manufacturing the luggage bag **10** includes providing the base **14**, and coupling a first wheel **136*a***, a second wheel **136*b***, a third wheel **142*a***, and a fourth wheel **142*b*** to the base **14**. The method further includes attaching or securing the back, first side, and second side portions **26**, **30**, **34** to the base **14**. In addition, the method includes attaching or securing the front portion **22** to the base **14**. The method also includes removably connecting the front portion **22** to the back, first side, and second side portions **26**, **30**, **34** by the connection member **98**. It should be appreciated that the disclosed method of manufacturing is illustrative, and the method may be completed in any suitable order or sequence of steps. In addition, two or more manufacturing steps may be completed concurrently.

[0146] The rolling luggage bag **10** provides advantages over known luggage in the art. Among them, utilizing an improved wheel arrangement of non-swivel wheels **136** combined with swivel wheels **142** that all remain in contact with the floor or other surface when the luggage bag **10** is in the upright position reduces the risk of unintended luggage movement while continuing to allow targeted rolling movement of the luggage bag **10** and reducing its overall weight. In addition, the positioning of the handle **74** also reduces user strain when the luggage bag **10** is rolled in a tilted position. Further, the user has unobstructed access to the interior chamber **102** defined by the luggage bag **10** through a butterfly opening, which additionally facilitates insertion and removal of large and/or bulky items. Moreover, the panels that define the front, back, first side, and second side portions **22**, **26**, **30**, **34** fold along a plurality of folds **42** that separate adjacent panels. By folding, the portions **22**, **26**, **30**, **34** cooperate to collapse into the base **14**, reducing the storage footprint of the luggage bag **10** when not in use.

EXAMPLE

[0147] Further described herein is a comparison between two luggage bags that utilized different collapsibility mechanisms. The results compared the effects that the panel construction and collapsibility mechanism had on the overall collapsibility of the bag. As discussed above, these variables can determine how much the golf bag is able to collapse for case of storage.

[0148] The control golf bag comprised twenty-two total panels arranged in five panel zones, similar to the luggage bag illustrated in FIGS. **1-6**. The panels of the control golf bag were arranged such that the side portions were permanently attached to the base. To open the control bag, the front portion was removed from the side portions, which pivoted or folded away from one another. Since the side portions were connected to the base, the control bag defined a butterfly opening, similar to that shown in FIGS. **8-9**. The control bag also included an angled fold to facilitate the collapsibility of the bag. To collapse the control bag, the front portion was removed from the side portions, folded along the folds between each panel, and received within the base. The side portions were then folded along the angled folds towards the base, similar to the luggage bag illustrated in FIG. **17**. In the collapsed position, the control bag was 43.8 cm wide, 33.6 cm deep, and 30.5 cm tall. In the upright position, the volume of the interior chamber of the control bag was 0.118 m^3.

[0149] The exemplary golf bag comprised twenty-one total panels arranged in five panel zones, similar to the luggage bag illustrated in FIGS. **27**A-**28**B. The panels of the exemplary bag were arranged such that the side portions were removable from the base, and the front portion included a sub-panel. To open the exemplary bag, the front portion was removed from the side portions, which folded away from one another. Since the side portions were removable from the base, the exemplary bag defined flat surface, similar to that shown in FIG. **29**. To collapse the exemplary bag, the front portion was removed from the side portions, the sub-panel was received within the base, and the remaining panels were folded and received within the base. The side portions were then folded inward toward the back portion and received within the base, similar to the luggage bag illustrated in FIGS. **31**A-**31**C. The exemplary bag included a clip that engaged with a loop to compress the cover into the base, similar to the luggage bag illustrated in FIG. **30**. In the collapsed position, the exemplary bag was 43.8 cm wide, 33.6 cm deep, and 21.6 cm tall. In the upright position, the volume of the interior chamber of the control bag was 0.118 m^3.

[0150] The comparison measured the volume of the interior chamber in the upright position and the volume and height of each golf bag in the collapsed position. The volume of the interior chamber is maximized for each golf bag to ensure that any golf bag can be retained within the interior chamber. The collapsed height and volume of the golf bag determines how easy it is to store the golf bag. The exemplary golf bag exemplified an ease of storage in comparison to the control bag, as discussed in further detail below.

TABLE 1

	Control Luggage Bag	Exemplary Luggage Bag
Interior Chamber Volume [m^3]	0.118	0.118
Collapsed Height [cm]	30.5	21.6
Collapsed Volume [cm^3]	0.0448	0.0317

[0151] Referring to Table 1, the exemplary golf bag demonstrated improved collapsibility over the control bag. While the bags had a similar interior chamber volume of 0.118 m^3 in the upright position, the exemplary bag demonstrated a smaller collapsed volume. Specifically, the control bag had a collapsed volume of 0.0448 m^3 and the exemplary bag had a collapsed volume of 0.0317 m^3 (29.2% smaller). Additionally, the exemplary bag was 8.9 cm shorter than the control bag in the collapsed position. The reduced height and volume of the exemplary luggage bag allows the bag to be stored more easily than the control bag. The improved performance of the exemplary golf bag was attributed to the panel construction and collapsibility mechanism.

[0152] As discussed above, the exemplary bag included side portions, which were fully detachable from the base. The side portions were then able to fully pivot inward toward the back portion before being received within the base. The exemplary bag further included a sub-panel on the front portion to help the front portion fit within the base. Together, the removable side portions and sub-panel optimized the storage capability of the base, resulting in a smaller collapsed height and volume. In contrast, the control bag included side portions that were permanently attached to the base, and the side portions were first folded along an angled fold before being received within the base. These angled folds did not optimize collapsibility, which led to the increase in the collapsed volume.

CLAUSES

[0153] Clause 1. A rolling luggage bag comprising: a cover fixedly coupled to a base, the base including a first side opposite a second side and a bottom face extending there between; wherein the cover comprises a plurality of panels, arranged in a direction away from the base, wherein the plurality of panels comprises a first panel zone proximate the base, a second panel zone adjacent the first panel zone in a direction away from the base, a third panel zone adjacent the second panel zone in a direction away from the base, a fourth panel zone adjacent the third panel zone in a direction away from the base, and a fifth panel zone adjacent the fourth panel zone in a direction away from the base; a first wheel and a second wheel coupled to the base, the first and second wheels configured to rotate about an axis of rotation and separated by a first distance along the axis of rotation, at least a portion of each of the first and second wheels projecting from the first side and from the bottom face; a third wheel and a fourth wheel coupled to the bottom face, the third and fourth wheels configured to independently swivel about a respective swivel axis and separated by a second distance extending between the swivel axes; wherein the plurality of panels defines an interior chamber, the interior chamber comprising an internal cover fixedly attached to the panels of the fifth panel zone.

[0154] Clause 2. The rolling luggage bag of clause 1, wherein the rolling luggage bag further comprises a plurality of rail members positioned in the second panel zone, and a plurality of rail extensions positioned on the base, wherein when the rolling luggage bag is in the upright position each rail member engages one of the pluralities of rail extensions to form a bumper that extends from the base along a portion of the cover.

[0155] Clause 3. The rolling luggage bag of clause 1, wherein the first distance is greater than the second distance.

[0156] Clause 4. The rolling luggage bag of clause 1, wherein the rolling luggage bag is configured to be movable along the surface through rotation of the first, second, third, and fourth wheels.

[0157] Clause 5. The rolling luggage bag of clause 1, wherein the rolling luggage bag is pivotable about the axis of rotation into a titled position such that the first and second wheels are in rolling contact with the surface, and the third and fourth wheels are removed from rolling contact with the surface, and wherein the rolling luggage bag is movable along the surface in the tilted position.

[0158] Clause 6. The rolling luggage bag of clause 1, wherein when the rolling luggage bag is in an upright position, the first, second, third, and fourth wheels all contact a surface the rolling luggage bag stands on.

[0159] Clause 7. The rolling luggage bag of clause 1, wherein the panels of the first panel zone are rectangular in shape and the panels of second panel zone are square in shape, wherein the square shape is defined by two separate triangular panels having an angled fold there between.

[0160] Clause 8. The rolling luggage bag of clause 1, wherein the panels of the third, fourth, and fifth panel zones are trapezoidal in shape.

[0161] Clause 9. The rolling luggage bag of clause 1, wherein the panels of the first panel zone have a greater stiffness than the panels of the second panel zone.

[0162] Clause 10. The rolling luggage bag of clause 1, wherein the first panel zone has a first handle on a side of the cover aligned with the first side of the base.

[0163] Clause 11. The rolling luggage bag of clause 1, wherein the second panel zone has a storage pocket.

[0164] Clause 12. The rolling luggage bag of clause 1, wherein the fourth panel zone has a plurality of second handles.

[0165] Clause 13. A collapsible luggage bag comprising: a back portion connected to a first side portion and a second side portion; a front flap removably connected to the first side portion, the second side portion, and the back portion by a connection member; a base connected to the back portion, the first side portion, the second side portion, and the front flap; wherein the base is fixedly connected to the back portion; wherein the back portion, first side portion, second side portion, and front flap comprise a plurality of panels, wherein the plurality of panels comprises a first panel zone proximate the base, a second panel zone adjacent the first panel zone in a direction away from the base, a third panel zone adjacent the second panel zone in a direction away from the base, a fourth panel zone adjacent the third panel zone in a direction away from the base, and a fifth panel zone adjacent the fourth panel zone in a direction away from the base, wherein each of the panel zones extends around a circumference defined by a portion of the back portion, first side portion, second side portion, and the front flap; wherein the second panel zone has a storage pocket, wherein the fourth panel zone has a plurality of second handles, wherein the first and second side portions pivot away from each other about respective folds between the respective side portion and the back portion when the front flap is at least partially removed; a first wheel and a second wheel coupled to the base, the first and second wheels configured to rotate about an axis of rotation and separated by a first distance along the axis of rotation, at least a portion of each of the first and second wheels projecting from the first side and from a bottom face; and a third wheel and a fourth wheel coupled to the bottom face, the third and fourth wheels configured to independently swivel about a respective swivel axis and separated by a second distance extending between the swivel axes; wherein the plurality of panels defines an interior chamber, the interior chamber comprising an internal cover fixedly attached to the panels of the fifth panel zone.

[0166] Clause 14. The collapsible luggage bag of clause 13, wherein the rolling luggage bag further comprises a plurality of rail members positioned in the second panel zone, and a plurality of rail extensions positioned on the base, wherein when the rolling luggage bag is in the upright position each rail member engages one of the pluralities of rail extensions to form a bumper that extends from the base along a portion of the cover.

[0167] Clause 15. The collapsible luggage bag of clause 13, wherein the second panel zone includes an arcuate portion of the front flap.

[0168] Clause 16. The collapsible luggage bag of clause 13, wherein when the rolling luggage bag is in an upright position, the first, second, third, and fourth wheels all contact a surface the rolling luggage bag stands on.

[0169] Clause 17. The collapsible luggage bag of clause 13, wherein the panels of the first panel zone are rectangular in shape, the panels of second panel zone are square in shape, wherein the square shape is defined by two separate triangular panels having an angled fold there between, and the panels of the third, fourth, and fifth panel zones are trapezoidal in shape.

[0170] Clause 18. The collapsible luggage bag of clause 13, wherein the panels of the first panel zone have a greater stiffness than the panels of the second panel zone.

[0171] Clause 19. The collapsible luggage bag of clause 13, wherein the connection member is a zipper, wherein the zipper includes at least two sliders.

[0172] Clause 20. The collapsible luggage bag of clause 13, wherein the collapsible luggage bag is pivotable about the axis of rotation into a titled position such that the first and second wheels are in rolling contact with the surface, and the third and fourth wheels are removed from rolling contact with the surface, and wherein the collapsible luggage bag is movable along the surface in the tilted position.

1. A rolling luggage bag comprising:

a cover fixedly coupled to a base, wherein the cover comprises a front portion, a back portion, a first side portion, and a second side portion opposite the first side portion, wherein the front portion, the back portion, the first side portion, and the second side portion are each formed from a plurality of panels, arranged in a direction away from the base and defining a plurality of folds between each of the plurality of panels, wherein;

the front portion is formed integrally with the base and is removably connected to the first side portion, the second side portion, and the base via a connection member;

the first side portion and the second side portions are formed integrally with back portion and are removably connected to the base via the connection member;

the first side portion and the second side portion pivot away from each other about respective folds between the respective side portion and the back portion when the front portion is at least partially removed;

the plurality of panels comprises a first panel zone proximate the base, a second panel zone adjacent the first panel zone in a direction away from the base, a third panel zone adjacent the second panel zone in a direction away from the base, a fourth panel zone adjacent the third panel zone in a direction away from the base, and a fifth panel zone adjacent the fourth panel zone in a direction away from the base;

the plurality of panels defines an interior chamber; and

the plurality of panels within each of the front portion, the back portion, the first side portion, and the second side portion vary in height such that the plurality of panes are offset relative to an adjacent panel zone.

2. The rolling luggage bag of claim **1**, wherein the base comprises a first side opposite a second side and a bottom face extending there between and a first wheel and a second wheel coupled to the base, the first and second wheels configured to rotate about an axis of rotation and separated by a first distance along the axis of rotation, at least a portion of each of the first and second wheels projecting from the first side and from the bottom face and a third wheel and a fourth wheel coupled to the bottom face, the third and fourth wheels configured to independently swivel about a respective swivel axis and separated by a second distance extending between the respective swivel axes.

3. The rolling luggage bag of claim **1**, wherein the connection member is a zipper, wherein the zipper includes at least two sliders.

4. The rolling luggage bag of claim **1**, wherein when the first and second side portions pivot away from each other, the first side portion, the second side portion, and the back portion define a flat surface.

5. The rolling luggage bag of claim **1**, wherein the rolling luggage bag further comprises a first inner pocket near the first side portion, and a second inner pocket near the second side portion.

6. The rolling luggage bag of claim **1**, wherein the first side portion defines a first bottom edge, and the second side portion defines a second bottom edge, and the first bottom edge and the second bottom edge each define a portion of a perimeter of the first and second side portions, respectively.

7. The rolling luggage bag of claim **6**, wherein a first portion of the connection member defines a portion of a perimeter of the front portion, and a second portion of the connection member defines a portion of a perimeter of the first and second side portions and extends along the first and second bottom edges.

8. The rolling luggage bag of claim **1**, wherein the rolling luggage bag further comprises a plurality of rail members positioned in the second panel zone, and a plurality of rail extensions positioned on the base, wherein when the rolling luggage bag is in an upright position each rail member engages one of the plurality of rail extensions to form a bumper that extends from the base along a portion of the cover.

9. The rolling luggage bag of claim **2**, wherein the rolling luggage bag is pivotable about the axis of rotation into a titled position such that the first and second wheels are in rolling contact with a surface, and the third and fourth wheels are removed from rolling contact with the surface, and wherein the rolling luggage bag is movable along the surface in a tilted position.

10. The rolling luggage bag of claim **2**, wherein when the rolling luggage bag is in an upright position, the first, second, third, and fourth wheels all contact a surface the rolling luggage bag stands on.

11. The rolling luggage bag of claim **1**, wherein the panels of the second panel zone are trapezoidal in shape.

12. The rolling luggage bag of claim **1**, wherein the panels of the first panel zone have a greater stiffness than the panels of the second panel zone.

13. A collapsible luggage bag comprising:

a cover fixedly coupled to a base, wherein the cover comprises a front portion, a back portion, a first side portion, and a second side portion opposite the first side portion, wherein the front portion, the back portion, the first side portion, and the second side portion are each formed from a plurality of panels, arranged in a direction away from the base and defining a plurality of folds between each of the plurality of panels, wherein;

the front portion is formed integrally with the base and is removably connected to the first side portion, the second side portion, and the base via a connection member;

the first side portion and the second side portions are formed integrally with back portion and are removably connected to the base via the connection member;

the first side portion and the second side portion pivot away from each other about respective folds between the respective side portion and the back portion when the front portion is at least partially removed;

the plurality of panels comprises a first panel zone proximate the base, a second panel zone adjacent the first panel zone in a direction away from the base, a third panel zone adjacent the second panel zone in a direction away from the base, a fourth panel zone adjacent the third panel zone in a direction away from the base, and a fifth panel zone adjacent the fourth panel zone in a direction away from the base;

the plurality of panels defines an interior chamber;

wherein the front portion of the second panel zone comprises a main front panel and a sub panel;

a sub fold defines a boundary between the main front panel and the sub panel; and

the sub panel raises a height of the main front panel such that the main front panel is located at an offset from each adjacent panel of the first and second side portions.

14. The collapsible luggage bag of claim **13**, wherein the base comprises a first side opposite a second side and a bottom face extending there between; a first wheel and a second wheel coupled to the base, the first and second wheels configured to rotate about an axis of rotation and separated by a first distance along the axis of rotation, at least a portion of each of the first and second wheels projecting from the first side and from the bottom face and a third wheel and a fourth wheel coupled to the bottom face, the third and fourth wheels configured to independently swivel about a respective swivel axis and separated by a second distance extending between the respective swivel axes.

15. The collapsible luggage bag of claim **13**, wherein the connection member is a zipper, wherein the zipper includes at least two sliders.

16. The collapsible luggage bag of claim **13**, wherein the panels of the second panel zone are trapezoidal in shape.

17. The collapsible luggage bag of claim **13**, wherein when the first and second side portions pivot away from each other, the first side portion, second side portion, and back portion define a flat surface.

18. The collapsible luggage bag of claim **13**, wherein the collapsible luggage bag further comprises a plurality of rail members positioned in the second panel zone, and a plurality of rail extensions positioned on the base, wherein when the collapsible luggage bag is in an upright position, each rail member engages one of the plurality of rail extensions to form a bumper that extends from the base along a portion of the cover.

19. The collapsible luggage bag of claim **14**, wherein the collapsible luggage bag is pivotable about the axis of rotation into a titled position such that the first and second wheels are in rolling contact with a surface, and the third and fourth wheels are removed from rolling contact with the surface, and wherein the collapsible luggage bag is movable along the surface in a tilted position.

20. The collapsible luggage bag of claim **14**, wherein when the collapsible luggage bag is in an upright position, the first, second, third, and fourth wheels all contact a surface the collapsible luggage bag stands on.

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